EE361 Electromechanical Energy Conversion I — METU EEE — full-semester study notes

Chapter V — Power Electronics

Güç Elektroniği · Weeks 13–14 of the 2026 syllabus

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- Lesson 28 — The step-down (buck) converter
- Lesson 29 — The step-up (boost) converter, regenerative braking and the two-quadrant chopper
- Lesson 30 — Single-phase diode rectifiers
- Lesson 31 — Three-phase diode rectifiers

> Assistant-written teaching notes (English with Turkish terminology). No teacher lecture deck for Chapter V is in the supplied set; the lessons follow the 2026 syllabus rows and Sen Ch. 10, with provenance stated in each lesson. Every numerical result is checked by `assistant/calculations/full-semester/run_all.py`; source pages are cited in each lesson. Printed 2026-11-02 from the Markdown sources in `assistant/lessons/`.

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Lesson 27 — Power semiconductor switches and the principles of switching power conversion

Chapter V: Power Electronics (Güç Elektroniği) · Week 13 · Lesson 27 of 31

Assistant-written teaching note. ***Provenance:*** *no teacher lecture deck for Chapter V is in the supplied set (the LN pack ends at DC machines, LN15). This lesson follows the 2026 syllabus rows "Power Semiconductor Switching Devices", "Principles of Switching Power Conversion" and "Applications of Power Electronics" and is built from the supplied textbook: Sen, Ch. 10 introduction and §10.1 (printed pp. 476–491, viewer 500–515), Fig. 10.1, Example 10.11 (printed pp. 545–547, viewer 569–571) and §10.9 (printed pp. 547–550, viewer 571–574). Loss-estimation formulas (§4) and the averaging/volt-second ideas (§5) are derived here from first principles; they are standard results, labelled as such. Numbers:* `lesson27_switching.py`.

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1. Place and goals

Every machine in Chapters II–IV is fed from somewhere. A DC motor needs an *adjustable* DC voltage (Lesson 26 §4.3), a battery charger needs DC from the AC grid, a PV panel needs its voltage raised to the grid level. Doing this with resistors wastes energy; doing it with **switches** wastes almost none. Power electronics (güç elektroniği) is the technology of converting electrical energy between forms — AC↔DC, one DC level to another — by switching semiconductor devices fully on and off.

After this lesson you should be able to:

1. explain why a switch converts power efficiently and a resistor or a linear transistor does not, and compute the efficiencies of the four regulators of Sen Example 10.11;
2. describe the external (terminal) behaviour of the diode, thyristor (SCR), triac, GTO/GCT, BJT, MOSFET and IGBT — what turns each on and off, voltage/current/frequency ranges, typical uses;
3. estimate conduction and switching losses of a switch from its data;
4. use the two tools of every converter analysis: the **average** of a switched waveform (duty cycle) and **periodic steady state** (inductor volt-second balance, capacitor charge balance);
5. place the converters of Lessons 28–31 in the family of AC–DC, DC–DC, DC–AC and AC–AC converters and name applications.

2. Why switch? Linear versus switching conversion

2.1 The problem with dissipative control

Consider getting 12 V across a 6 Ω load from a 24 V source (Sen Example 10.11). A series resistor of 6 Ω does it — and burns as much power as the load: efficiency V_o/V_{in} = 50 %. A linear (series-pass) transistor is just an adjustable resistor: the same 50 %. Generally, any element that carries the load current I while dropping a voltage $V_{in} - V_o$ dissipates $(V_{in} - V_o)I$, so

$$\eta_{linear} \leq \frac{V_o}{V_{in}}. \tag{27.1}$$

The same happens in machines: the old way to control a subway car's series DC motor was a series resistor (Sen Fig. 10.1a); the energy you did not want in the motor became heat in the resistor.

2.2 The ideal switch

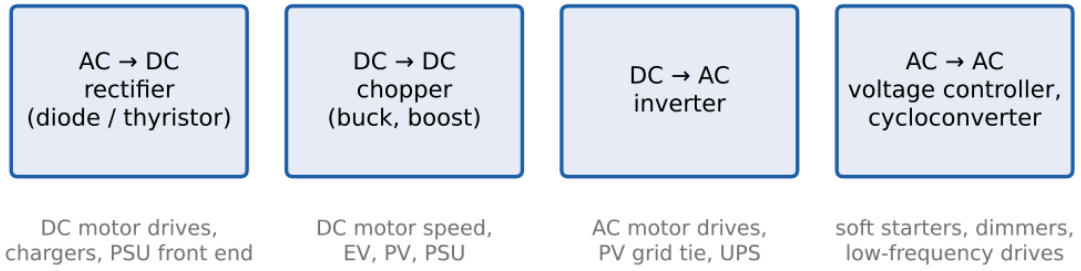
An **ideal switch** is either **on** ($v = 0$, any current) or **off** ($i = 0$, any voltage). In both states $p = vi = 0$. If the switch alternates quickly, the load sees a *chopped* voltage: V_{in} for a time t_{on}, zero for the rest of the period T. Its average is

$$V_o = \frac{t_{on}}{T} V_{in} = D\,V_{in}, \qquad D = \frac{t_{on}}{T} \text{ (duty cycle, görev oranı)} \tag{27.2}$$

(Sen Eq. 10.18–10.19 writes the duty ratio as α). In Example 10.11(d) the 6 Ω resistor receives V_{in} half the time: average voltage 12 V, input power = output power = DV_{in}^2/R = 48 W, efficiency 100 % (the load sees an rms voltage of $\sqrt{D}\,V_{in}$ = 17 V because the waveform is pulsed — Lesson 28 adds an L–C filter so the load gets a smooth 12 V).

The idea of switching conversion: control the *time* the source is connected, not the voltage drop across a dissipating element. Then filter the pulsed waveform with energy-storage elements (L, C), which are lossless in principle. The price: switching harmonics and electromagnetic interference (radio-frequency interference, RFI — Sen §10.1), and the need for filters.

Power electronic converters (Sen Ch. 10 introduction); EE361 syllabus: AC→DC diode rectifiers and DC→DC buck/boost



2.3 Families of converters (Sen Ch. 10 introduction)

Converter	Conversion	Typical device	Use (Sen)
Rectifier (doğrultucu), diode or controlled	AC → DC (fixed or variable)	diodes, thyristors	DC motor drives, chargers, supply front ends
Chopper (kıyıcı), DC–DC converter	DC → DC (variable)	MOSFET, IGBT, GTO	DC motor speed control, power supplies
Inverter (evirici)	DC → AC (variable f, V)	IGBT, MOSFET	AC motor drives, PV, UPS
AC voltage controller	AC → AC (variable V)	triac, thyristors	soft starting, dimmers
Cycloconverter	AC → AC (lower f)	thyristors	large low-speed drives

The 2026 syllabus asks for the first two families in their basic forms: **diode rectifiers** (single- and three-phase, Lessons 30–31) and **buck and boost DC–DC converters** (Lessons 28–29).

3. Power semiconductor switches (Sen §10.1)

Real switches differ from the ideal one in four ways: a small **on-state voltage** or resistance, a small **off-state leakage**, finite **switching times**, and **limits** of voltage, current, di/dt and dv/dt. They also differ in *how they are controlled*:

- **uncontrolled** — the circuit decides: diode;
- **semi-controlled** — turned on by the gate, off only by the circuit: thyristor (SCR), triac;
- **fully controlled** — turned on and off by the control terminal: GTO/GCT, BJT, MOSFET, IGBT.

3.1 Diode (diyot) — Sen §10.1.8

Two-layer pn device. Reverse biased: open circuit (until reverse breakdown). Forward biased: conducts with 0.8–1 V drop. It turns on and off by itself, following the circuit. After forward conduction it carries a brief **reverse-recovery** current while stored charge is removed (1–5 µs for conventional diodes, a few hundred ns for fast-recovery types) — important at high switching frequency. Ratings up to about 4000 V, 2000 A. In this chapter diodes are the rectifying elements (Lessons 30–31) and the freewheeling/output diodes of choppers (Lessons 28–29).

3.2 Thyristor (SCR, tristör) — Sen §10.1.1

Four-layer pnpn device with anode, cathode and **gate**. With forward voltage it blocks until a gate current pulse (typically 20–200 mA for 10–50 µs) switches it into conduction (1–2.5 V drop). **Once on, the gate has no control**: it turns off only when its current falls below the holding current — naturally at an AC zero crossing (*natural/line commutation*) or by a commutation circuit that reverse-biases it (*forced commutation*). It must then stay reverse-biased for the turn-off time t_{off} (10–100 µs) or it re-conducts. Protection: series inductance limits di/dt, an R–C snubber limits dv/dt (Sen Fig. 10.6). Highest ratings of all devices (Sen Table 10.2 lists a 12 kV, 1500 A unit); low switching loss; slow. Used in controlled rectifiers (Sen §10.2), HVDC.

3.3 Triac, GTO and GCT — Sen §10.1.2–10.1.4

- **Triac:** two antiparallel SCRs in one device, gated for either polarity; low-power AC control (blenders, vacuum cleaners, dimmers); lower dv/dt capability.
- **GTO (gate turn-off thyristor):** turned on like an SCR and **off by a large negative gate current** (anode-to-gate current ratio only 3–5: a 1000 A GTO needs ≈ 250 A of negative gate current). Higher switching losses → used below about 1 kHz; no external commutation circuit needed.
- **GCT/IGCT:** GTO-like structure with much faster turn-off and lower losses; ratings to about 6000 V, 6000 A; has replaced the GTO in high-power industrial converters.

3.4 Power transistors: BJT, MOSFET, IGBT — Sen §10.1.5–10.1.7

	BJT	MOSFET	IGBT
Control	base **current** (gain as low as 10; Darlington for more)	gate **voltage** (≈ no steady gate current; a few A during transitions to charge gate capacitance)	gate **voltage** (MOSFET-like gate)
On state	≈ constant drop (saturation)	**resistance** $R_{DS(on)}$: loss $\propto I^2$	≈ constant drop (≈ 2–3 V) like a BJT

	BJT	MOSFET	IGBT
Speed	µs	≈ 50 ns; up to MHz	µs-range (like BJT)
Weak point	secondary breakdown (needs SOA care, snubbers)	high $R_{DS(on)}$ at high voltage	slower than MOSFET
Typical domain	largely replaced	low voltage, high frequency, ≤ a few kW	600–3300 V, kW–MW drives (preferred between about 600 and 2000 V)
Reverse blocking	no (antiparallel diode used)	no; **integral body diode**	no (antiparallel diode in modules)

Sen's numerical illustration of the MOSFET trade-off (p. 488): a 600 V device carrying 40 A with $R_{DS(on)}$ = 0.1 Ω drops 4 V (160 W of conduction loss), whereas a device for below about 50 V can have $R_{DS(on)} < 10$ mΩ (0.4 V, 16 W at 40 A). That is why MOSFETs dominate low-voltage, high-frequency supplies and IGBTs dominate medium-voltage motor drives.

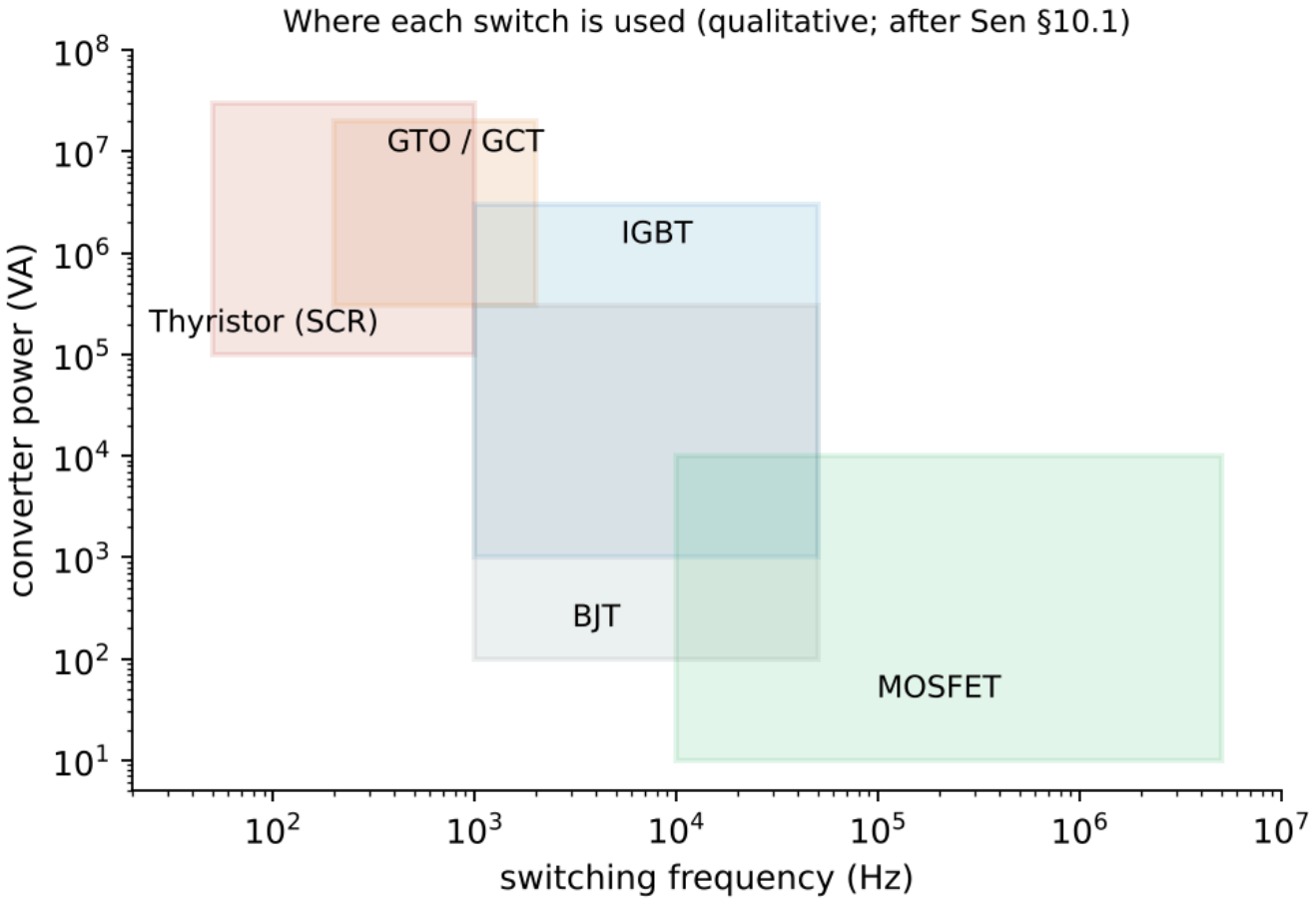


Figure 27.2 is a qualitative map assembled from the ratings and frequency remarks in Sen §10.1 (e.g. GTOs at or below 1 kHz, power transistors to about 50 kHz, MOSFETs to MHz); boundaries are indicative only.

3.5 Safe operating area and snubbers

Each transistor has a **safe operating area (SOA)** bounded by peak current, power dissipation, (for BJTs) secondary breakdown and peak voltage (Sen Fig. 10.12). Simultaneous high voltage and high current during turn-off is the danger; a **polarized snubber** (diode–capacitor–resistor)

diverts current into a capacitor so that the voltage rises only after the current has fallen (Sen Fig. 10.13). MOSFETs and IGBTs have essentially no secondary breakdown, but still need protection against voltage spikes.

4. Losses in a real switch

4.1 Conduction loss (iletim kaybı)

While on, a switch dissipates

$$P_{cond} = V_{on}\, I_{avg} \ \text{(constant-drop devices: diode, IGBT, BJT, SCR)}, \qquad P_{cond} = R_{on}\, I_{rms}^2 \ \text{(MOSFET)} \tag{27.3}$$

For a switch that carries a flat current I for a fraction D of the period: $I_{avg} = DI$, $I_{rms} = \sqrt{D}\, I$.

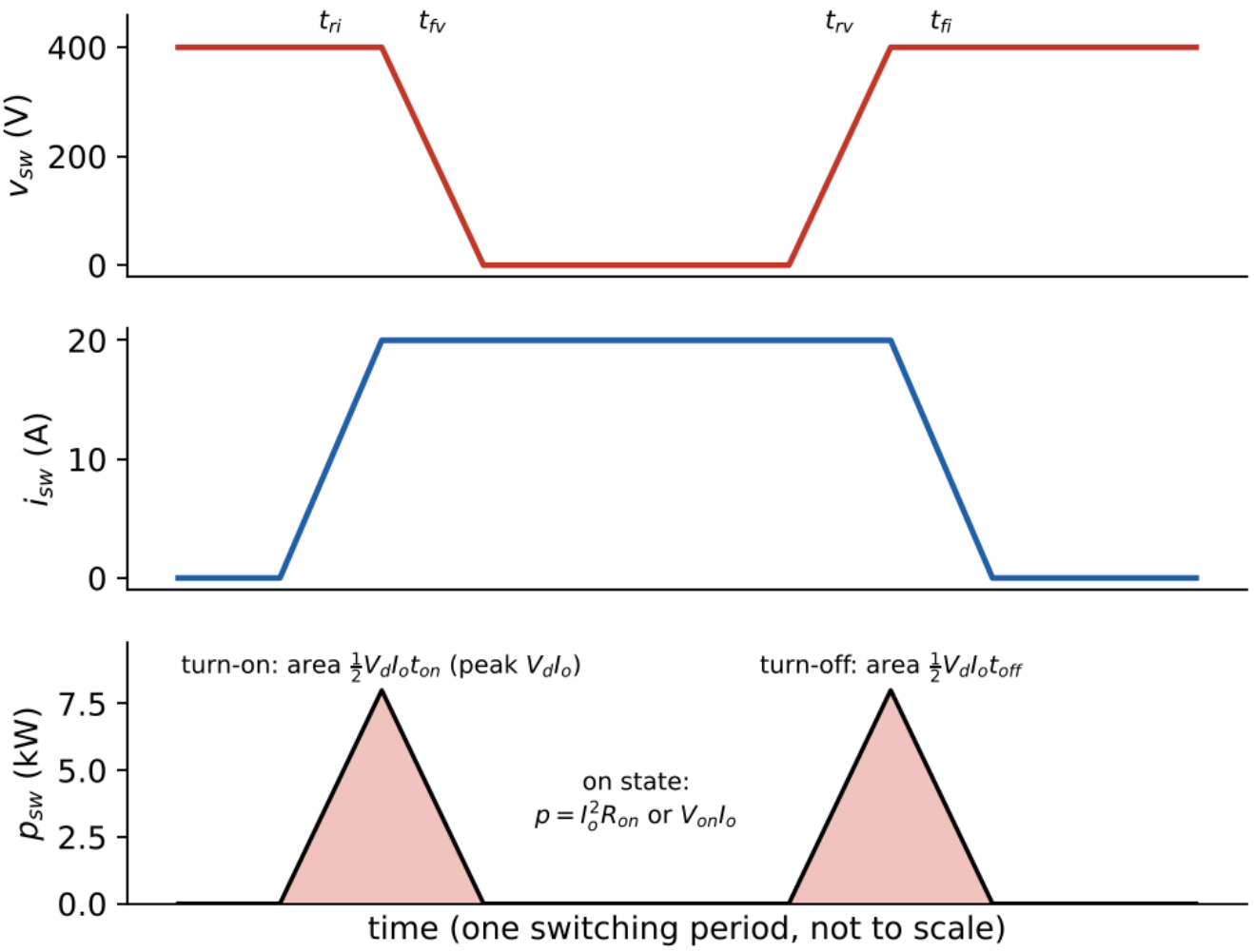
4.2 Switching loss (anahtarlama kaybı)

During each transition voltage and current overlap. For the usual converter situation — an inductor keeps the load current I_o constant, and a diode clamps the switch voltage to V_d (a *clamped inductive* switching) — the turn-on proceeds in two steps (Figure 27.3): the current rises from 0 to I_o while the voltage stays at V_d (the diode still conducts), then the voltage falls to zero at full current. Each step is a triangle of height $V_d I_o$:

$$E_{on} = \tfrac{1}{2} V_d I_o\, t_{ri} + \tfrac{1}{2} V_d I_o\, t_{fv} = \tfrac{1}{2} V_d I_o\, t_{on}, \qquad E_{off} = \tfrac{1}{2} V_d I_o\, t_{off},$$

$$\boxed{P_{sw} = \tfrac{1}{2} V_d I_o \left(t_{on} + t_{off}\right) f_s} \tag{27.4}$$

(a standard idealized estimate with linear transitions; `lesson27_switching.py` integrates $v\,i$ numerically to confirm it). Switching loss grows **linearly with switching frequency** — the fundamental trade-off: higher f_s shrinks inductors and capacitors (Lessons 28–29) but costs switching loss. Thyristors switch with low loss (low current when the voltage is high, Sen p. 480); GTOs have high turn-off loss and stay below about 1 kHz.



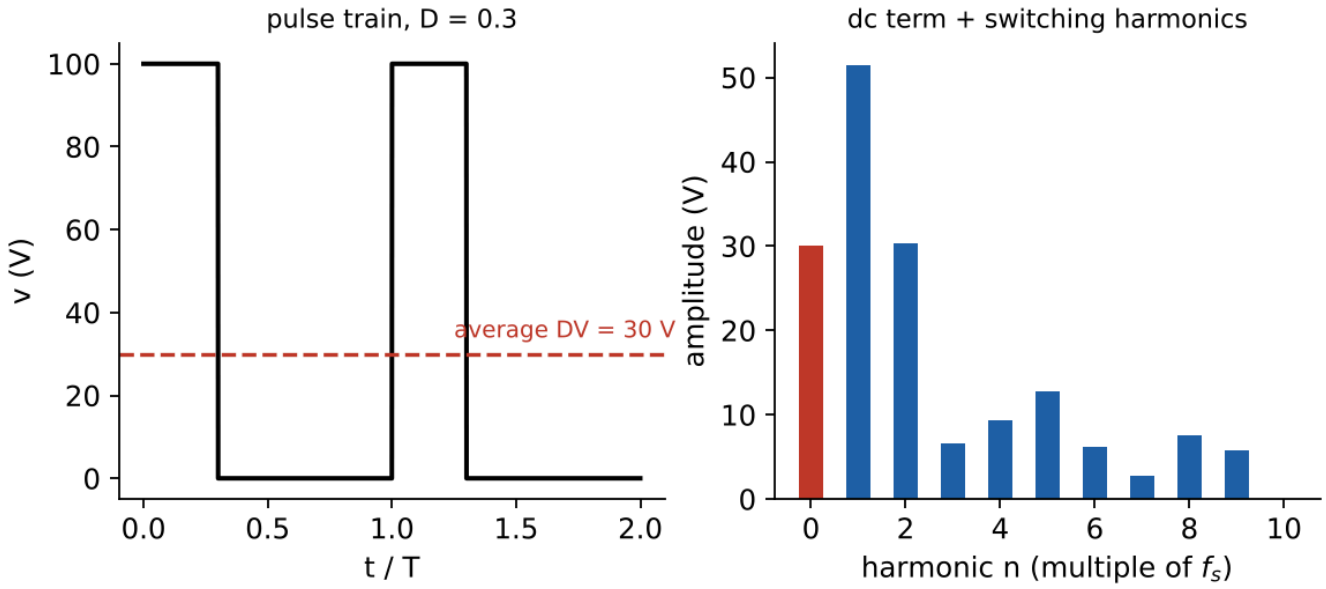
5. Tools for analysing switching converters

5.1 Average value and duty cycle

A pulse train of height V and duty cycle D has

$$\text{average } DV, \qquad \text{rms } \sqrt{D}\,V, \qquad \text{harmonic } n \text{ amplitude } \frac{2V}{n\pi}\,\lvert \sin(n\pi D) \rvert \ \text{ at } n f_s. \tag{27.5}$$

The average carries the useful power; the harmonics at multiples of f_s are what the filter must remove. An L–C low-pass filter with corner $f_0 \ll f_s$ attenuates the first harmonic by roughly $(f_0/f_s)^2$ (Figure 27.4). The higher f_s, the smaller the L and C needed for the same attenuation.



5.2 Periodic steady state: volt-second and charge balance

In steady state every waveform repeats each period. For an inductor, $v_L = L\,di_L/dt$ integrated over one period gives $L[i_L(T) - i_L(0)] = \int_0^T v_L\,dt$. Since $i_L(T) = i_L(0)$:

$$\boxed{\int_0^T v_L\,dt = 0 \quad\Longleftrightarrow\quad \langle v_L \rangle = 0} \qquad \text{(inductor volt-second balance).} \tag{27.6}$$

Likewise for a capacitor, $\int_0^T i_C\,dt = C[v_C(T) - v_C(0)] = 0$: **the average capacitor current is zero** (charge balance). Sen uses exactly these facts: “there is no average voltage across the inductor” (Eq. 10.37), “no average current in the capacitor” (Eq. 10.42), and the inductor-current rise during t_{on} equals its fall during t_{off} (boost, Eqs. 10.20a–c). They also explain Lesson 26: an armature inductance cannot sustain an average voltage, so $V_t = E_a + I_a R_a$ in averages even with a chopped v_t (Sen Eq. 10.4).

Small-ripple approximation. If L and C are large enough that the ripple is small compared with the averages, treat inductor currents and capacitor voltages as constant when computing the *other* quantities, then compute the ripple from the slopes. This is how Lessons 28–29 derive every formula.

5.3 Power balance

With ideal (lossless) switches, L and C: $P_{in} = P_{out}$. Combined with $V_o = DV_{in}$ this gives the “DC transformer” view of the buck converter: $I_{in}/I_o = V_o/V_{in} = D$ (Sen Eq. 10.41, Fig. 10.55e).

6. Applications (Sen Ch. 10 introduction, §10.9)

- **DC motor drives:** chopper control replacing series resistors (Sen Fig. 10.1); converter-fed motors (Sen Examples 4.11–4.13; Lesson 26 Example 26.9; Lessons 28–29).
- **Electric vehicles and regenerative braking:** bidirectional choppers (KERS of a Formula-1 car, Final Q4 2019; e-scooter, Jan 2022 Final Q4; mining truck, Final 2018 Q3).
- **Computer power supplies (Sen §10.9, Fig. 10.56):** 120 V AC → diode bridge + boost with power-factor correction → 400 V DC → isolated DC–DC → 12 V → buck on the motherboard → ≈ 1 V for the CPU. Every stage is one of the circuits of this chapter.
- **Renewables and grids:** PV boost converters, inverters, HVDC.

7. Methods and common mistakes

1. **“A transistor in its active region is a switch.”** No — in the active region it is a dissipating resistor. Power converters keep devices fully on or fully off.
2. **Duty cycle vs firing angle.** Choppers use $D = t_{on}/T$ (Sen calls it α); controlled rectifiers use the firing angle α (Lesson 30 mentions it only for comparison). Do not mix the two αs.
3. **MOSFET loss uses rms current; diode/IGBT loss uses average current** (constant drop). Using the average for a MOSFET underestimates loss.
4. **Volt-second balance applies to averages in steady state only.** During a transient (start-up) the inductor current changes from period to period.
5. **Units of switching times:** ns for MOSFETs, µs for IGBTs/thyristors. A factor 1000 error in t_{sw} is a factor 1000 in switching loss.

8. Worked examples

Example 27.1 — Four ways to make 12 V from 24 V (Sen Example 10.11)

Given. 24 V source, 12 V required across 6 Ω (2 A). **Find** the efficiency of (a) a resistive divider, (b) a zener regulator with 0.1 A zener current, (c) a linear transistor regulator, (d) a switching regulator. **Solution.** (a) Series resistor $(24-12)/2$ = 6 Ω: $\eta = 12/24$ = **50 %**. (b) Supply current 2.1 A, series resistor $12/2.1$ = 5.71 Ω, $P_{in} = 24 \times 2.1$ = 50.4 W, P_o = 24 W: η = **47.6 %**. (c) V_{CE} = 12 V at ≈ 2 A: $\eta \approx V_o/V_{in}$ = **50 %**. (d) $D = 12/24$ = 0.5; the load receives DV_{in}^2/R = 48 W and the source supplies 48 W: η = **100 %** with an ideal switch. **Interpretation.** Only the switching circuit beats (27.1). With a real MOSFET a few watts are lost (Example 27.3), still above 90 %.

Example 27.2 — Resistor versus chopper speed control of a DC motor (original, data of Sen Example 4.13)

Given. Separately excited motor on a 120 V supply, R_a = 0.5 Ω, $K_a\Phi$ = 0.05 V/rpm, constant-torque load requiring I_a = 20 A. Run it at 1000 rpm (a) with a series resistor, (b) with a chopper. **Solution.** At 1000 rpm E_a = 50 V, so the armature needs $V_t = E_a + I_aR_a$ = 60 V. (a) Series resistor $(120-60)/20$ = **3 Ω**, dissipating 1200 W; input 120×20 = 2400 W; output E_aI_a = 1000 W ⇒ η = **41.7 %**. (b) Chopper with $D = 60/120$ = 0.5: average source current DI_a = 10 A, input 1200 W, output 1000 W, only the 200 W in R_a is lost ⇒ η = **83.3 %** (ideal switch). **Interpretation.** Same speed, same torque, half the input power. This is Sen Fig. 10.1’s point.

Example 27.3 — Conduction losses (Sen p. 488 data and original buck case)

(a) Sen’s comparison at 40 A: 600 V MOSFET (0.1 Ω) → 4 V drop, **160 W**; low-voltage MOSFET (10 mΩ) → 0.4 V, **16 W**; an IGBT with 2.7 V on-state drop (the IGBT column of Sen Table 10.2) → **108 W**. **(b)** In a buck converter at D = 0.5 carrying a flat 20 A: the MOSFET conducts half the time, $I_{rms} = \sqrt{0.5} \times 20$ = 14.1 A ⇒ $P = 0.1 \times 14.1^2$ = **20 W**; the freewheeling diode carries 20 A for the other half, I_{avg} = 10 A ⇒ $P = 0.9 \times 10$ = **9 W**.

Example 27.4 — Switching loss and the choice of frequency (original)

Given. Clamped inductive switching, V_d = 400 V, I_o = 20 A. A MOSFET with $t_{on} + t_{off}$ = 100 ns; an IGBT with $t_{on} + t_{off}$ = 1 µs. **Find** P_{sw} at 50 kHz for both, and for the IGBT at 10 kHz. **Solution.** (27.4): MOSFET $\frac{1}{2}(400)(20)(100 \times 10^{-9})(50 \times 10^3)$ = **20 W**; IGBT at 50 kHz **200 W**; IGBT at 10 kHz **40 W**. **Interpretation.** The IGBT’s slower transitions make it a 10–20 kHz device at this power; a MOSFET can run much faster — if its conduction loss at 400 V is acceptable (Example 27.3).

Example 27.5 — Volt-second balance (original)

Given. An inductor (100 µH) in a periodic circuit sees +40 V for 3 µs, then $-V$ for 2 µs. **Find** V for periodic steady state and the current ripple. **Solution.** (27.6): $40 \times 3 = V \times 2 \Rightarrow V$ **= 60 V**. Ripple $\Delta i = 40 \times 3 \times 10^{-6}/100 \times 10^{-6}$ **= 1.2 A** (same rise and fall). If V were smaller, the current would rise a little more each period than it falls — not a steady state (the simulation in the verifier shows zero drift only for 60 V).

Example 27.6 — Average, rms and harmonics of a pulse train (original)

Given. 100 V pulses, D = 0.3. **Find** average, rms and the first harmonics. **Solution.** Average 30 V, rms $\sqrt{0.3} \times 100$ = 54.8 V. Harmonic amplitudes $\frac{200}{n\pi}|\sin(0.3n\pi)|$: 51.5, 30.3, 6.6, 9.4, 12.7 V for n = 1…5 (FFT agrees; Figure 27.4). The fundamental (51.5 V) is *larger* than the useful average: without a filter the load sees mostly AC. An L–C filter with corner 1 kHz at f_s = 20 kHz reduces it by about $(1/20)^2$, to ≈ 0.13 V.

9. Optional practice

P27.1. A 48 V battery must supply 5 V at 2 A to electronics. Compare a linear regulator with an ideal buck converter.

▼ **Solution**
Linear: $\eta \le 5/48$ = 10.4 %; it dissipates $(48-5) \times 2$ = 86 W to deliver 10 W. Ideal buck: $D = 5/48$ = 0.104, η = 100 %, input current $D \times 2$ = 0.21 A. This is why linear regulators are used only for small voltage differences or tiny currents.

P27.2. A buck converter switches 400 V and carries 20 A at D = 0.5 and 50 kHz (output 200 V, 4 kW). The MOSFET has R_{on} = 0.1 Ω and $t_{on} + t_{off}$ = 100 ns; the diode drop is 1 V. Estimate the total device loss and efficiency.

▼ **Solution**
MOSFET conduction $0.5 \times 20^2 \times 0.1$ = 20 W; switching (27.4) 20 W; diode $0.5 \times 20 \times 1$ = 10 W. Total 50 W ⇒ $\eta = 4000/4050$ = 98.8 %.

P27.3 (concept). Why can a thyristor not be used as the switch of a DC chopper without extra circuitry, while it works naturally in an AC rectifier?

▼ **Solution**
A thyristor turns off only when its current falls to zero. In an AC circuit the supply voltage reverses every half cycle and does this automatically (line commutation). In a DC chopper the current never falls to zero by itself, so a forced-commutation circuit (a precharged capacitor that reverse-biases the SCR, Sen Fig. 10.35) is needed — or a fully controlled device (GTO, IGBT, MOSFET) is used instead.

P27.4 (concept). A designer doubles the switching frequency of a converter. What happens to (a) inductor current ripple for the same L, (b) switching loss, (c) conduction loss?

▼ **Solution**

a. Ripple ∝ T (Lesson 28), so it halves (or L can be halved for the same ripple). (b) Switching loss doubles (27.4). (c) Conduction loss is unchanged to first order (it depends on duty cycle and current, not on frequency).

10. Summary

Idea	Key relation
Linear/dissipative regulation	$\eta \leq V_o/V_{in}$
Ideal switch	$p = vi = 0$ on and off; $V_o = DV_{in}$
Device control	diode: none; SCR/triac: on by gate, off by circuit; GTO/GCT/BJT/MOSFET/IGBT: on and off by control
Conduction loss	$V_{on}I_{avg}$ (diode, IGBT, SCR) or $R_{on}I_{rms}^2$ (MOSFET)
Switching loss	$\frac{1}{2}V_dI_o(t_{on}+t_{off})f_s$
Pulse train	avg DV, rms $\sqrt{D}V$, harmonics $\frac{2V}{n\pi}\lvert\sin n\pi D\rvert$
Steady state	$\langle v_L\rangle = 0$, $\langle i_C\rangle = 0$; ideal converter $P_{in} = P_{out}$

11. Vocabulary (EN–TR)

English	Türkçe
power electronics	güç elektroniği
switch / switching	anahtar / anahtarlama
duty cycle (duty ratio)	görev oranı (doluluk oranı)
switching frequency	anahtarlama frekansı
diode / thyristor / triac	diyot / tristör / triyak
gate (control terminal)	kapı (geyt)
conduction loss / switching loss	iletim kaybı / anahtarlama kaybı
on-state voltage drop / on-resistance	iletim gerilim düşümü / iletim direnci
reverse recovery	ters toparlanma
safe operating area (SOA)	güvenli çalışma bölgesi
snubber circuit	bastırma (snubber) devresi
natural / forced commutation	doğal / zorlamalı komütasyon
rectifier / chopper / inverter	doğrultucu / kıyıcı / evirici
converter	dönüştürücü (çevirici)
average / rms value	ortalama / etkin (rms) değer
harmonic, ripple	harmonik, dalgalanma
volt-second balance	volt-saniye dengesi

12. Sources

- Sen, *Principles of Electric Machines and Power Electronics*, 3rd ed.: Ch. 10 introduction and Fig. 10.1 (printed p. 476, viewer 500); §10.1.1–10.1.8, Figs. 10.2–10.16, Tables 10.1–10.2 (printed pp. 477–491, viewer 501–515); §10.4.1 Eqs. 10.18–10.19 (printed p. 515, viewer 539); §10.8 Eqs. 10.37–10.42 and Example 10.11 (printed pp. 542–547, viewer 566–571); §10.9 (printed pp. 547–550, viewer 571–574).
- Syllabus `sources/syllabus/EE361_syllabus_2026_ver_5.pdf` p. 5 (Week 13 rows). No Chapter V teacher deck supplied.
- Exam context: Final Q4 2019 (KERS), Jan 2022 Final Q4 (scooter), Final 2018 Q3 (eDumper) — solved in Lessons 28–29 and the Chapter IV collection.

- Verification: `lesson27_switching.py` (11 checks: efficiencies, rms/average currents, numerical switching-energy integration, volt-second steady state, FFT harmonics and Parseval).

Lesson 28 — The step-down (buck) converter

Chapter V: Power Electronics (Güç Elektroniği) · Week 14 · Lesson 28 of 31

Assistant-written teaching note. ***Provenance:*** *no teacher deck for Chapter V is supplied. Built from Sen §10.4.1 (step-down chopper, printed pp. 515–517, viewer 539–541), §10.8 (buck power supply, printed pp. 542–547, viewer 566–571), Example 10.5(a) (printed pp. 521–523, viewer 545–547) and Example 4.13 (chopper-fed DC motor, printed pp. 186–188, viewer 210–212). The output-voltage ripple, the CCM/DCM boundary and the DCM voltage ratio (§4–5) are derived here from first principles (standard results, beyond Sen's text). Exam problems: Final Q4 2019 (KERS, with annotated solution), Jan 2022 Final Q4(a)–(d) (e-scooter, with key), Jan 2023 Final Q4(d) (field supply, with key). Numbers and switching simulations:* `lesson28_buck.py`.

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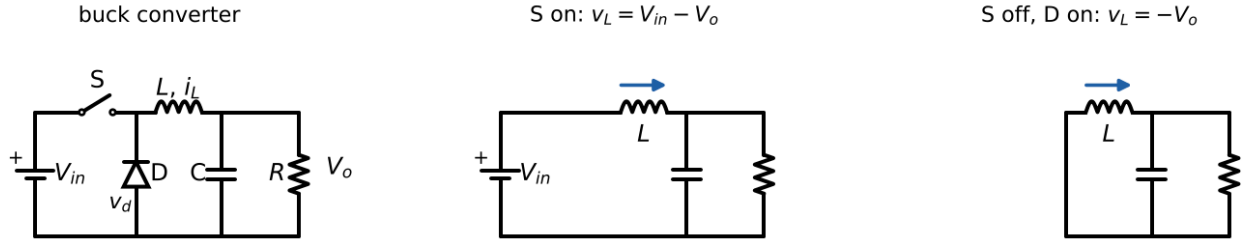
1. Place and goals

The buck converter is the switching circuit that produces a *lower*, adjustable DC voltage from a fixed DC source. It is the chopper that drives DC motors (Sen §10.4.1, Lesson 26), the regulator on every computer motherboard (Sen §10.8–10.9), and the "controllable voltage source" assumed in the DC-drive exam problems of Chapter IV.

After this lesson you should be able to:

1. draw the circuit and its two switching states; sketch v_d, v_L, i_L, i_C, v_o and the input current;
2. derive $V_o = DV_{in}$ from inductor volt-second balance and $I_{in} = DI_o$ from power balance;
3. compute the inductor-current ripple, the peak and valley currents, the output-voltage ripple, and choose L and C for given ripple limits;
4. find the boundary between continuous and discontinuous conduction and the output voltage in discontinuous conduction;
5. analyse a chopper-fed DC motor, where the armature inductance is the filter inductor and the back EMF is the "output voltage", including starting with current limitation and regenerative energy flow.

2. Circuit and operation



The switch S (MOSFET or IGBT) connects the input V_{in} to the node d for $t_{on} = DT$ in each period $T = 1/f_s$. An inductor L and capacitor C filter the pulsed node voltage v_d; a diode D (the **freewheeling diode**, serbest geçiş diyodu) gives the inductor current a path when S is off.

- **S on,** $0 < t < DT$ (Sen Fig. 10.55b): $v_d = V_{in}$, the diode is reverse biased, $v_L = V_{in} - V_o > 0$, and i_L rises linearly (Sen Eq. 10.35): $i_L = I_1 + \dfrac{V_{in} - V_o}{L} t$.

- **S off,** $DT < t < T$ (Sen Fig. 10.55c): the inductor keeps its current flowing, forcing the diode on; $v_d = 0$, $v_L = -V_o$, and i_L falls linearly (Sen Eq. 10.36): $i_L = I_2 - \dfrac{V_o}{L}(t - DT)$.

Assumption for §3: the capacitor is large enough that $v_o \approx V_o$ is constant (small-ripple approximation), and the inductor current never reaches zero — **continuous conduction mode (CCM)**.

3. Steady-state relations in CCM

3.1 Voltage conversion ratio

Volt-second balance on L (Lesson 27, Eq. 27.6): $(V_{in} - V_o)DT + (-V_o)(1 - D)T = 0$, hence

$$\boxed{V_o = DV_{in}} \qquad (0 \le D \le 1;\ \text{Sen Eq. } 10.37). \tag{28.1}$$

Equivalently: the inductor has no average voltage, so V_o equals the average of v_d, which is DV_{in} (Sen's argument). The output is independent of the load in CCM — the converter behaves like a controllable voltage source.

3.2 Currents and power

The capacitor has zero average current, so the average inductor current equals the load current (Sen Eq. 10.42): $I_L = I_o = V_o/R$. The source supplies current only while S is on, so its average is $I_{in} = DI_L = DI_o$. With ideal components $P_{in} = V_{in}I_{in} = DV_{in}I_o = V_oI_o = P_o$ (Sen Eqs. 10.39–10.41): a **DC transformer** with turns ratio D.

Device currents (flat-current approximation): switch $I_{S,avg} = DI_o$, $I_{S,rms} = \sqrt{D}\,I_o$; diode $I_{D,avg} = (1 - D)I_o$. Both devices must block V_{in}.

3.3 Inductor-current ripple

From the off-interval slope (Sen Eq. 10.43):

$$\boxed{\Delta I_L = I_2 - I_1 = \frac{V_o(1 - D)T}{L} = \frac{V_{in}D(1 - D)}{Lf_s}}, \qquad I_{1,2} = I_o \mp \tfrac{1}{2}\Delta I_L \tag{28.2}$$

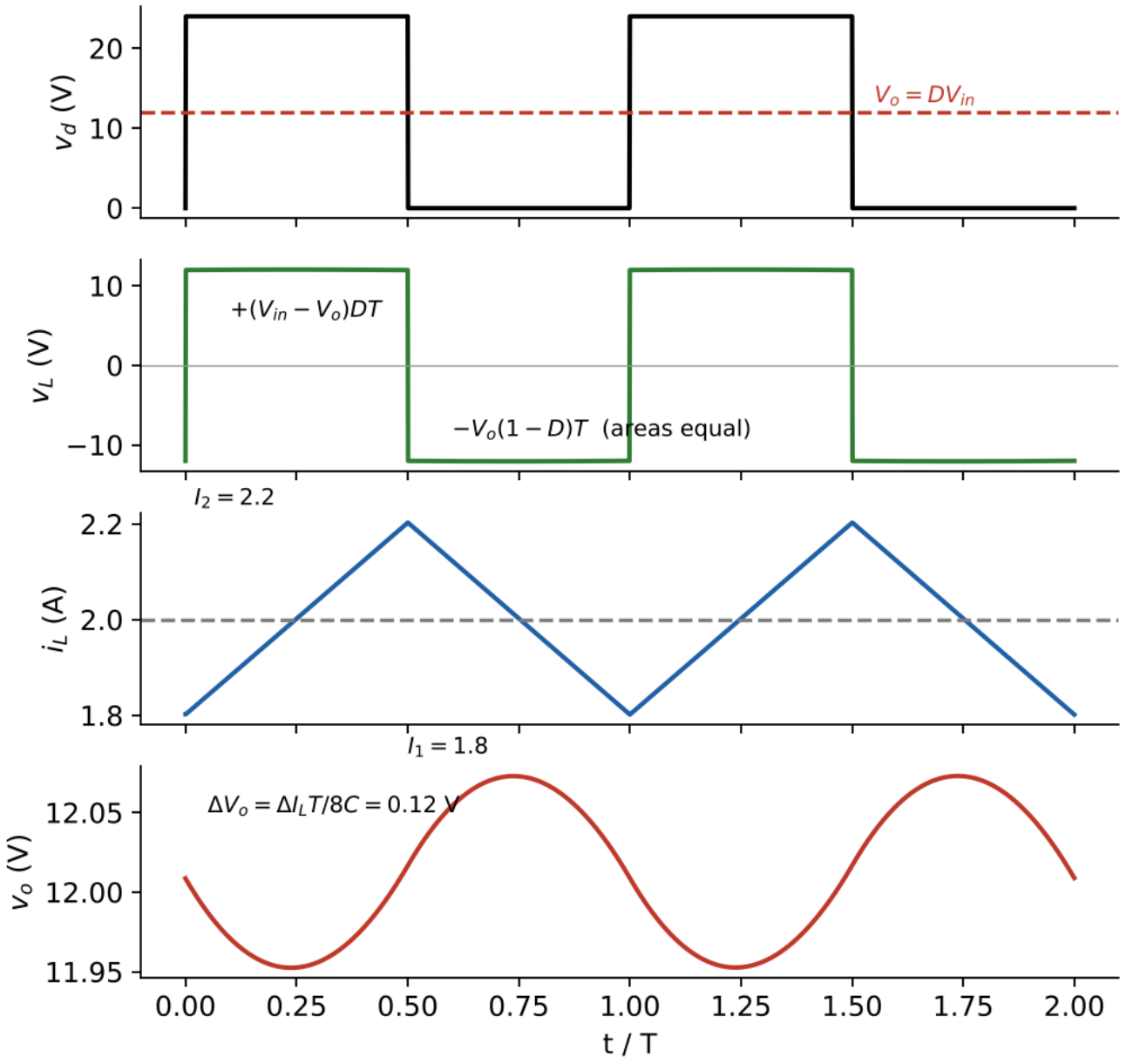
(the on-interval slope gives the same result, as it must). For fixed V_{in} the ripple is largest at $D = 0.5$: $\Delta I_{L,max} = V_{in}/(4Lf_s)$. Peak switch current I_2 sets the device and inductor-saturation rating.

3.4 Output-voltage ripple

The capacitor takes the AC part of i_L (the triangle), the load takes the DC part. The capacitor current is positive while $i_L > I_o$ — for half a period, during which the charge delivered is the area of a triangle of base $T/2$ and height $\Delta I_L/2$:

$$\Delta Q = \tfrac{1}{2} \cdot \frac{T}{2} \cdot \frac{\Delta I_L}{2} = \frac{\Delta I_L T}{8}, \qquad \boxed{\Delta V_o = \frac{\Delta Q}{C} = \frac{\Delta I_L}{8Cf_s} = \frac{V_o(1 - D)}{8LCf_s^2}}. \tag{28.3}$$

Ripple falls with the *square* of switching frequency — why supplies switch at hundreds of kHz (Sen §10.9). (Derived here; Sen §10.8 assumes a constant output voltage.)



4. Continuous and discontinuous conduction

As the load current decreases, the valley current $I_1 = I_o - \Delta I_L/2$ decreases; at $I_1 = 0$ the converter is on the **boundary** of continuous conduction:

$$I_{o,B} = \frac{\Delta I_L}{2} = \frac{V_o(1-D)T}{2L} \qquad \Longleftrightarrow \qquad L_{crit} = \frac{(1-D)RT}{2}. \tag{28.4}$$

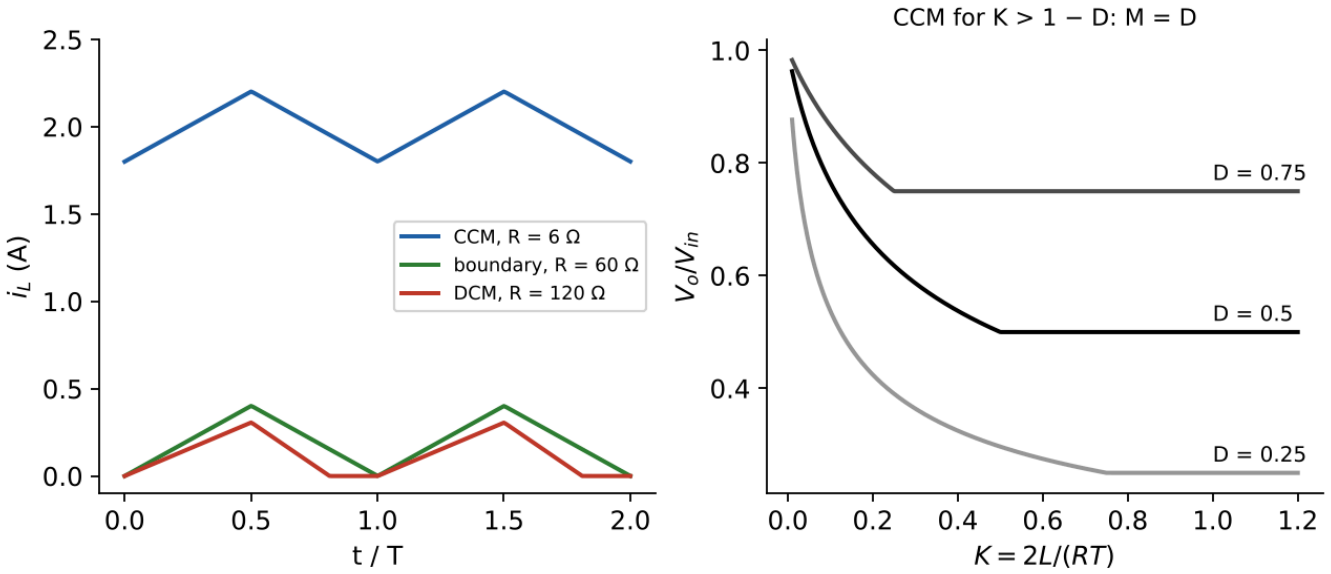
For $I_o < I_{o,B}$ (or $L < L_{crit}$), the current would try to go negative, but the diode blocks it: i_L reaches zero before the end of the period and stays there. This is **discontinuous conduction mode (DCM)**. In DCM the diode conducts for $D_2 T$ only; volt-second balance gives $(V_{in} - V_o)D = V_o D_2$, and the average inductor current (the area of one triangle per period) must equal V_o/R:

$$\frac{\Delta I_L}{2}(D + D_2) = \frac{V_o}{R}, \qquad \Delta I_L = \frac{(V_{in} - V_o)DT}{L}.$$

Eliminating D_2 and writing $K = 2L/(RT)$:

$$\boxed{\frac{V_o}{V_{in}} = \frac{2}{1 + \sqrt{1 + 4K/D^2}} \quad (\text{DCM},\ K < 1 - D)}, \qquad \frac{V_o}{V_{in}} = D \quad (\text{CCM},\ K \geq 1 - D). \tag{28.5}$$

In DCM the output depends on the load and rises above DV_{in} at light load — a closed-loop controller then reduces D. The boundary $K = 1 - D$ is (28.4) rewritten. (Derivation ours; confirmed by simulation in Example 28.1.)



5. The buck converter as a DC motor drive

In a chopper-fed motor (Sen Figs. 4.62, 10.34) there is no separate L or C: the **armature inductance** L_a is the filter inductor and the back EMF E_a plays the role of the output voltage. The average armature equation (an inductor sustains no average voltage, Sen Eq. 10.4):

$$\boxed{V_t = DV_s = E_a + I_a R_a}. \tag{28.6}$$

- **Speed control:** at constant load torque (I_a fixed), speed is set by D (Sen Example 4.13).
- **Current ripple:** with R_a small, $L_a\, di/dt \approx V_s - V_t$ during t_{on}, so $\Delta I_a \approx V_s D(1-D)T/L_a$ — the same as (28.2). The torque ripple is $K_a \Phi\, \Delta I_a$.
- **Current-limited start:** at standstill $E_a = 0$; connecting the full supply would give V_s/R_a, typically 10 × rated. The chopper starts with $D = I_{rated} R_a / V_s$ and raises D as the speed (and E_a) builds up (Jan 2022 Final Q4(c)–(d)).
- **Direction of power flow:** a simple buck passes power only from source to motor ($I_a > 0$, quadrant I). Regenerative braking needs the boost connection or a two-quadrant chopper (Lesson 29). The “bidirectional variable voltage supply” in the KERS problem is such a converter.

6. Methods and common mistakes

1. **Start with volt-second balance**, not with formulas: write v_L in each interval, set the average to zero.
2. **Do not forget that $I_L = I_o$ for the buck** (the inductor is in series with the load) — but $I_{in} = DI_o$.
3. **Ripple uses $(1-D)$ and the output voltage** in the off interval ($\Delta I = V_o(1-D)T/L$) or $(V_{in} - V_o)DT/L$ in the on interval; mixing them gives wrong answers.
4. $\Delta V_o = \Delta I_L/(8Cf_s)$ **assumes the capacitor takes the whole ripple current** (ideal capacitor, no ESR). The 8 comes from a triangle of base $T/2$ and height $\Delta I/2$.
5. **DCM**: $V_o = DV_{in}$ is no longer valid; check I_o against $I_{o,B}$ first.

6. **Motor drives**: use $V_t = E_a + I_a R_a$ with averages; “steady state” means constant speed (constant E_a) as well as periodic current.

7. Worked examples

Example 28.1 — Design of a 24 V → 12 V buck converter (original, the operating point of Sen Example 10.11)

Given. V_{in} = 24 V, V_o = 12 V, R = 6 Ω, f_s = 50 kHz. Inductor ripple ≤ 20 % of I_o, output ripple ≤ 1 % of V_o. **Find** D, L, C, $I_{1,2}$, the CCM boundary, device currents; then (e) the output if the load becomes 120 Ω at the same D. **Solution.**

a. $D = 12/24 = 0.5$; I_o = 2 A; $I_{in} = DI_o$ = 1 A.
b. ΔI_L = 0.4 A ⇒ $L = V_o(1-D)T/\Delta I_L = 12(0.5)(20\times10^{-6})/0.4$ = **300 µH**; I_1 = 1.8 A, I_2 = 2.2 A.
c. ΔV_o = 0.12 V ⇒ $C = \Delta I_L T/(8\Delta V_o) = 0.4(20\times10^{-6})/(8\times0.12)$ = **8.33 µF** (choose 10 µF).
d. Boundary: $I_{o,B} = \Delta I_L/2$ = 0.2 A, i.e. R_B = 60 Ω; $L_{crit} = (1-D)RT/2$ = 30 µH (our L is 10 × critical at full load). Switch: avg 1 A, rms $\sqrt{0.5}\times 2$ = 1.41 A, peak 2.2 A, blocks 24 V; diode avg 1 A.
e. R = 120 Ω: $I_o \approx$ 0.1 A < 0.2 A ⇒ DCM. $K = 2L/(RT) = 2(300\times10^{-6})/(120\times20\times10^{-6})$ = 0.25 ⇒ $V_o/V_{in} = 2/(1+\sqrt{1+4(0.25)/0.25}) = 2/(1+\sqrt{5})$ = 0.618 ⇒ V_o = **14.83 V**, not 12 V.
Verification. The switching simulation (Figure 28.2) gives V_o = 12.006 V, ΔI_L = 0.401 A, ΔV_o = 0.120 V; in DCM it settles at 14.84 V with i_L reaching zero each period (Figure 28.3).

Example 28.2 — Speed range of a chopper-fed DC motor (Sen Example 4.13)

Given. 120 V DC supply, R_a = 0.5 Ω, L_a = 20 mH, $K_a\Phi$ = 0.05 V/rpm; constant-torque load needing I_a = 20 A (continuous). **Find** (a) the speed range, (b) the duty-cycle range; (c, our extension) the current ripple at D = 0.5 if the chopper runs at 1 kHz. **Solution.** (a, b) Zero speed: E_a = 0 ⇒ $V_t = I_a R_a$ = 10 V ⇒ $D = 10/120 = 1/12$. Maximum at D = 1: $E_a = 120 - 10$ = 110 V ⇒ N = 110/0.05 = **2200 rpm**. Range: 0 < N < 2200 rpm for **1/12 < D < 1** (Sen’s answer).

c. At D = 0.5: V_t = 60 V, E_a = 50 V, N = 1000 rpm; $\Delta I_a \approx V_s D(1-D)/(L_a f) = 120(0.25)/(0.02\times1000)$ = **1.5 A** (7.5 % of 20 A) — the worst case over D. Simulation: 1.499 A about a 20.0 A average.

Example 28.3 — Two-quadrant chopper, motoring (Sen Example 10.5(a))

Given. 120 V supply, f_c = 250 Hz (T = 4 ms), $K\Phi$ = 0.1 V/rpm, R_a = 0.2 Ω, ripple-free current; S1 and D1 operate (buck), n = 400 rpm, I_o = 100 A. **Find** t_{on} and the powers. **Solution.** $V_0 = E_a + I_a R_a = 0.1(400) + 100(0.2)$ = 60 V ⇒ $t_{on}/T = 60/120 = 0.5$ ⇒ t_{on} = **2 ms**. Motor developed power $E_a I_o$ = **4000 W**; R_a loss $100^2(0.2)$ = **2000 W**; source power $V(i_s)_{avg} = 120\times100\times0.5$ = **6000 W** = 4000 + 2000 ✓. The source current is a 100 A pulse train of duty 0.5. (Part (b), regeneration, is Lesson 29 Example 29.2.)

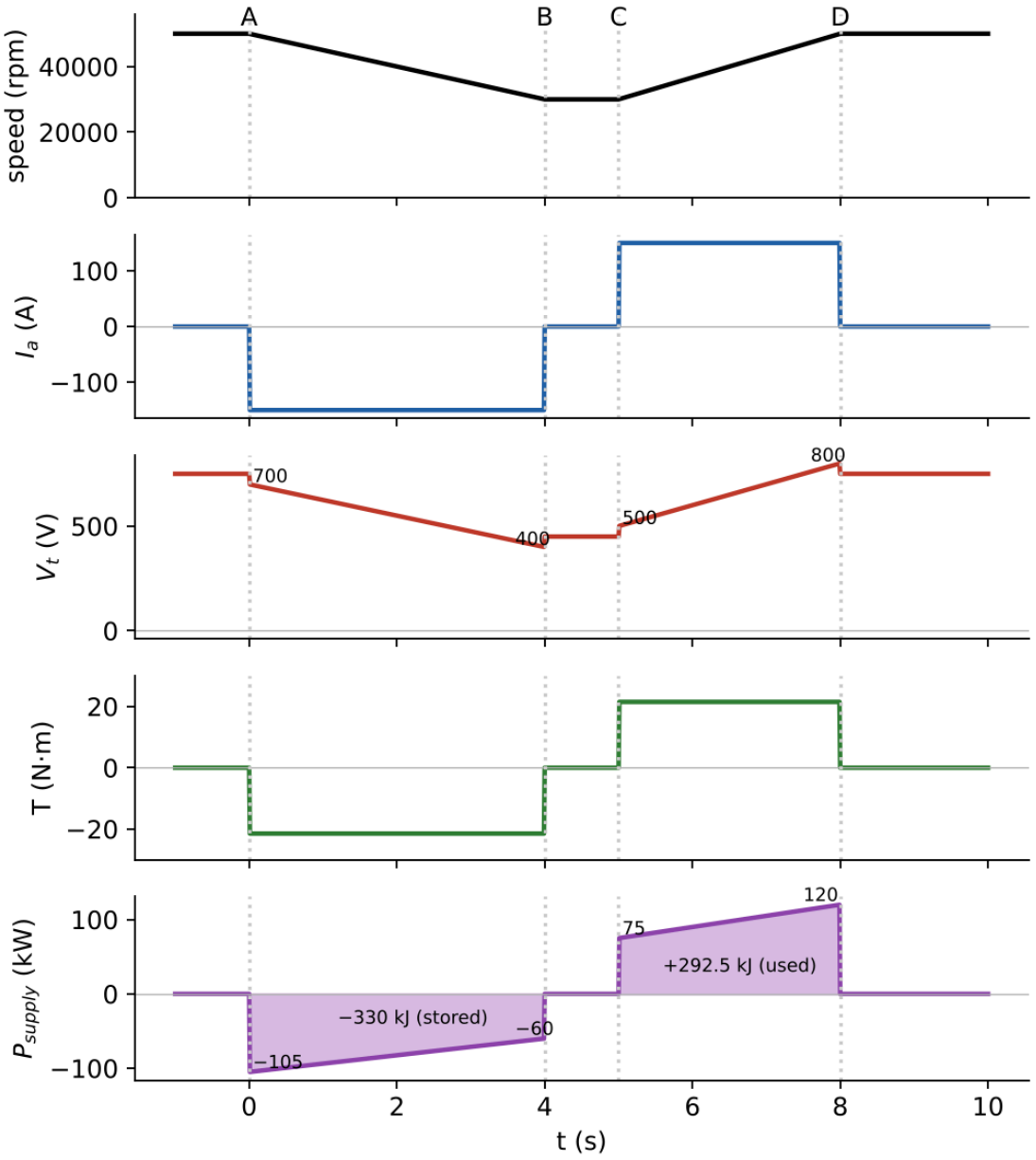
Example 28.4 — KERS motor-generator unit (Final Q4 2019, Final Recitation 2020)

Source. `exams/archive/Final/Final Recitation 2020 - Final Q4 2019.pdf` pp. 1–6 and `… - Solution of Final Q4 2019 - Annotated.pdf` pp. 1–4. **Key confirmed. Given.** PM DC machine (constant flux) as the MGU-K: E_a = 750 V at 50 000 rpm; R_a = 1/3 Ω. Speed profile: 50 000 rpm → (A to B, 4 s, linear) → 30 000 rpm → hold → (C to D, 3 s, linear) → 50 000 rpm; I_a = −150 A during A–B, +150 A during C–D, zero otherwise. Supply = ideal bidirectional variable-voltage

source. **(a)** $K_a\Phi = 750/(50\,000 \times 2\pi/60)$ **= 0.1432 V·s/rad**; motoring at 150 A: $T = K_a\Phi I_a$ **= 21.5 N·m**. **(b)** $V_t = E_a + I_aR_a = 750 + 50$ **= 800 V**; $\eta = E_aI_a/V_tI_a$ = 750/800 **= 93.8 %**. **(c)(i)** With $V_t = E_a + I_aR_a$ and $P_s = V_tI_a$ at the corners (Figure 28.4):

Instant	E_a **(V)**	I_a **(A)**	V_t **(V)**	**T (N·m)**	P_s **(kW)**
before A	750	0	750	0	0
A⁺	750	−150	700	−21.5	−105
B⁻	450	−150	400	−21.5	−60
B–C	450	0	450	0	0
C⁺	450	+150	500	+21.5	+75
D⁻	750	+150	800	+21.5	+120
after D	750	0	750	0	0

Between the corners E_a varies linearly with speed, so V_t and P_s are linear ramps. **(c)(ii)** Energy = area under $P_s(t)$: $W_{AB} = \frac{1}{2}(-105 - 60)(4)$ **= −330 kJ** (stored in the energy store), $W_{CD} = \frac{1}{2}(75 + 120)(3)$ **= 292.5 kJ** (used). Energy bookkeeping (ours): during A–B the machine converts $\bar{E}_aI_at = 600 \times 150 \times 4$ = 360 kJ of mechanical energy, of which $I_a^2R_at$ = 30 kJ is lost in R_a → 330 kJ stored. During C–D it needs 270 kJ mechanical + 22.5 kJ loss = 292.5 kJ. **(d)** Buck converter from V_s = 800 V at 2 kHz, D = 0.75, R_a ignored, L_a = 30 mH. (i) $v_L = V_s - E_a$ during DT_s, $-E_a$ during $(1 - D)T_s$; i_a is a triangle around its average. In steady state $E_a = DV_s$ = 600 V (40 000 rpm). (ii) $\Delta i_a = (V_s - E_a)DT_s/L_a = 200(0.75)(0.5 \times 10^{-3})/0.03$ **= 2.5 A**, tiny compared with the 150 A rated current (simulation: 2.50 A) — the armature inductance alone is an adequate filter at 2 kHz.



Example 28.5 — Supplying a field winding from 800 V (Jan 2023 Final Q4(d))

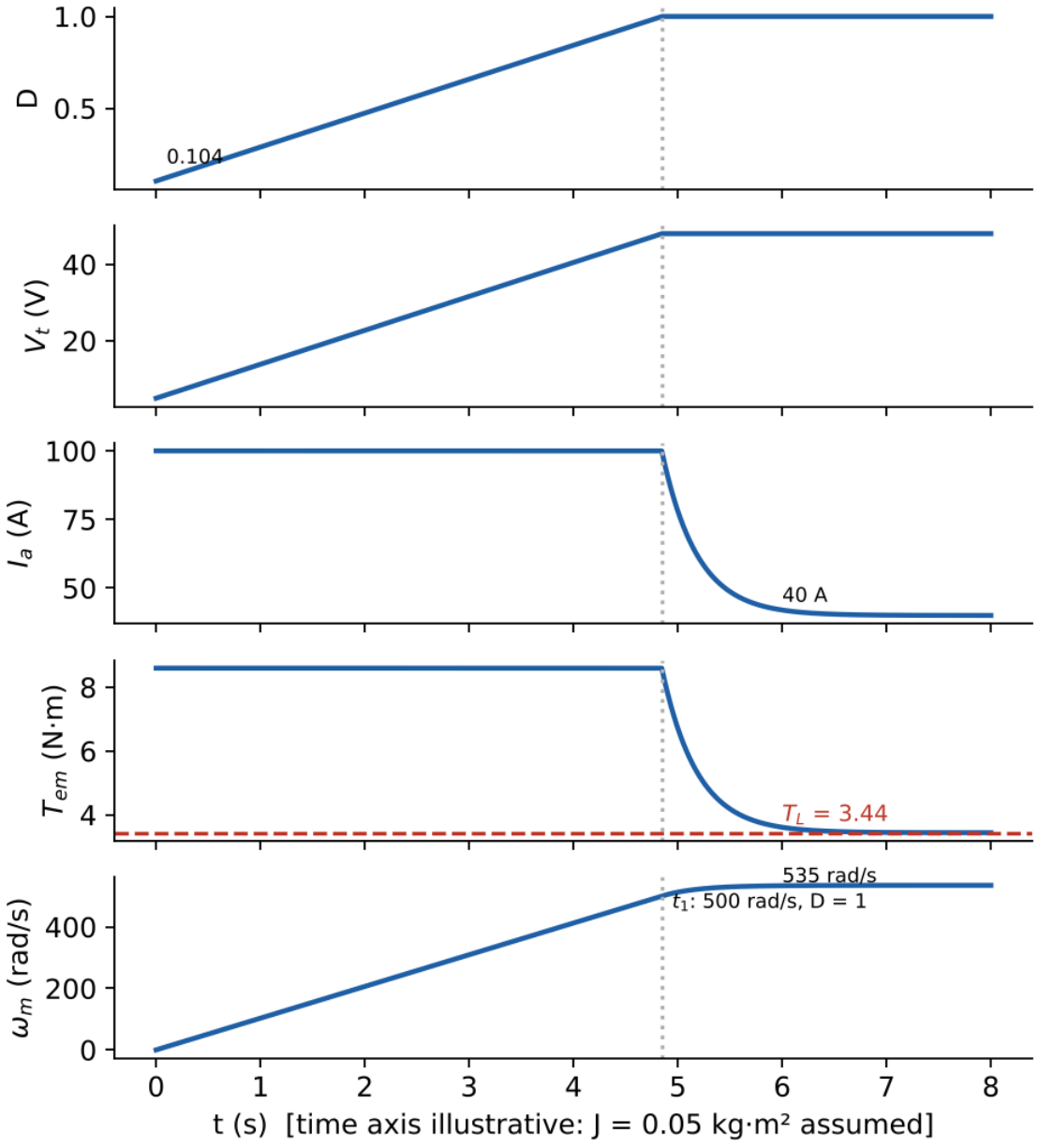
Source. `exams/archive/20221/EE361_Final_2022_Q4_Solution.pdf` p. 14 (16 January 2023 final). **Key confirmed.** Parts (a)–(c) of this crane-motor question are in the Chapter IV collection §10. **Given.** Rated field current 1.5 A, R_f = 200 Ω, main supply 800 V. **Solution.** Required field voltage $V_f = 1.5 \times 200 = 300$ V < 800 V ⇒ a buck converter (switch from the 800 V rail, freewheeling diode across the field; the field inductance L_f is the filter inductor, a capacitor is optional). $D = 300/800 =$ **0.375**. The source current is a pulse train: ≈ 1.5 A while S is on, zero while the diode freewheels; its average is $DI_f =$ **0.5625 A**, which matches the power balance $800 \times 0.5625 = 300 \times 1.5 = 450$ W. **Remark (key).** A buck converter gives only positive field current; part (c) of the question needs a reversed field, which requires an H-bridge (four switches) in practice.

Example 28.6 — Current-limited start of an e-scooter (Jan 2022 Final Q4(a)–(d))

Source. `exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf` pp. 11–13 (31 January 2022 final). **Key confirmed. Given.** PM DC motor treated as separately excited: rated 4300 W output, 4775 rpm, 100 A; battery 48 V; rotational losses and inductance ignored. **(a)** $\omega_r = 4775 \times$

$2\pi/60$ = 500.0 rad/s. $E_a = P/I_a$ = 43 V; $R_a = (48 - 43)/100$ = **0.05 Ω**; $K_a\Phi = 43/500$ = **0.086 V·s/rad**; $T = K_a\Phi I_a$ = **8.6 N·m** (check $T\omega$ = 4300 W ✓); $\eta = 4300/(48 \times 100)$ = **89.6 %**. **(b)** Direct start: $E_a = 0 \Rightarrow I = 48/0.05$ = **960 A** (9.6 × rated), T = **82.6 N·m**. **(c)** Buck between battery and motor, current limited to 100 A at standstill: $V_t = I_a R_a$ = **5 V** ⇒ $D = 5/48$ = **0.104**. **(d)** Uphill load torque 3.44 N·m (40 % of rated). Fastest acceleration without exceeding rated current:

- *Phase 1 (constant current, 0 < t < t_1):* hold I_a = 100 A, so T_{em} = 8.6 N·m > 3.44 N·m and the speed rises linearly. To keep I_a constant as E_a grows, raise $V_t = E_a + 5$ V, i.e. D increases linearly from 0.104. This lasts until D = 1, V_t = 48 V, E_a = 43 V: ω = **500 rad/s** (rated speed).
- *Phase 2 ($t > t_1$, D = 1):* V_t stays 48 V; as speed rises E_a rises, the current falls, torque falls, acceleration decreases — an exponential approach with mechanical time constant $JR_a/(K_a\Phi)^2$.
- *Steady state:* $T_{em} = T_L \Rightarrow I_a = 3.44/0.086$ = **40 A**, $E_a = 48 - 40(0.05)$ = 46 V ⇒ ω = 46/0.086 = **535 rad/s** (above rated speed, as the hint allows). The inertia is not given, so the time axis cannot be numbered; Figure 28.5 uses an assumed J = 0.05 kg·m² only to draw the shapes (t_1 = 4.8 s, τ_m = 0.34 s in that illustration). Part (e) (boost for double speed) is Lesson 29 Example 29.3.



8. Optional practice

P28.1 (design). Design a 48 V → 12 V buck converter for 5 A at 100 kHz with $\Delta I_L \le 1$ A and $\Delta V_o \le 50$ mV. Give D, L, C and the load current at the CCM boundary.

▼ **Solution**
$D = 0.25$. $L \ge V_o(1-D)T/\Delta I_L = 12(0.75)(10^{-5})/1$ = **90 µH**. $C \ge \Delta I_L T/(8\Delta V_o) = 10^{-5}/(0.4)$ = **25 µF**. Boundary $I_{o,B} = \Delta I_L/2$ = **0.5 A**.

P28.2 (DCM). The converter of P28.1 (L = 90 µH, D held at 0.25) supplies a 48 Ω load. Find V_o.

▼ **Solution**
$I_o \approx 0.25$ A < 0.5 A ⇒ DCM. $K = 2L/(RT) = 2(90\times10^{-6})/(48\times10^{-5}) = 0.375$; $4K/D^2 = 24$; $V_o/V_{in} = 2/(1+5) = 1/3$ ⇒ V_o = **16 V** (not 12 V). The voltage control loop must lower D at light load.

P28.3 (KERS variant). In Example 28.4(d), what is the ripple if the switching frequency is raised to 10 kHz? At what duty cycle is the ripple at 2 kHz largest, and how large?

▼ **Solution**
Ripple $\propto 1/f_s$: **0.5 A** at 10 kHz. Maximum at $D = 0.5$: $V_s/(4L_a f_s) = 800/(4\times0.03\times2000)$ = **3.33 A**.

P28.4 (concept). Why does a buck converter feeding a DC motor usually need no output capacitor, whereas a buck power supply does?

▼ **Solution**
The motor's back EMF changes only with speed, which the rotor inertia keeps essentially constant over a switching period: E_a acts like a constant voltage "capacitor", and L_a limits the current ripple. A resistive or electronic load has no such internal voltage; without C the load voltage would be the pulsed v_d (or Ri_L with large ripple).

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9. Summary

Quantity (CCM, ideal)	Buck
Voltage ratio	$V_o = DV_{in}$
Currents	$I_L = I_o$, $I_{in} = DI_o$; switch rms $\sqrt{D}I_o$, diode avg $(1-D)I_o$
Inductor ripple	$\Delta I_L = V_o(1-D)T/L$ (max $V_{in}/4Lf_s$ at D = 0.5)
Output ripple	$\Delta V_o = \Delta I_L/(8Cf_s)$
CCM boundary	$I_{o,B} = \Delta I_L/2$; $L_{crit} = (1-D)RT/2$
DCM ratio	$V_o/V_{in} = 2/(1+\sqrt{1+4K/D^2})$, $K = 2L/RT$
Motor drive	$DV_s = E_a + I_aR_a$; start with $D = I_{lim}R_a/V_s$

10. Vocabulary (EN–TR)

English	Türkçe
step-down (buck) converter	düşürücü (buck) dönüştürücü
chopper	kıyıcı
freewheeling diode	serbest geçiş diyodu

English	Türkçe
continuous / discontinuous conduction mode	sürekli / süreksiz iletim kipi
inductor current ripple	endüktans akımı dalgalanması
output voltage ripple	çıkış gerilimi dalgalanması
critical inductance	kritik endüktans
duty cycle	görev oranı
current limiting	akım sınırlama
armature inductance	endüvi endüktansı
power supply	güç kaynağı
regenerative braking	rejeneratif (geri kazanımlı) frenleme

11. Sources

- Sen 3rd ed.: §10.4.1, Eqs. 10.18–10.19, Fig. 10.34 (printed pp. 515–517, viewer 539–541); §10.4.4 and Example 10.5 (printed pp. 521–523, viewer 545–547); §10.8, Eqs. 10.35–10.45, Fig. 10.55 (printed pp. 542–545, viewer 566–569); Example 10.11 (printed pp. 545–547); §4.5 choppers, Eqs. 4.62–4.63, Fig. 4.62, Example 4.13 (printed pp. 186–188, viewer 210–212).
- Final Recitation 2020 — Final Q4 2019 (question pp. 1–6) and annotated solution (pp. 1–4).
- Jan 2022 Final Q4(a)–(d): `exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf` pp. 11–13 (key).
- Jan 2023 Final Q4(d): `exams/archive/20221/EE361_Final_2022_Q4_Solution.pdf` p. 14 (key).
- Derived here (standard, beyond Sen's text): Eq. 28.3 output ripple, Eqs. 28.4–28.5 boundary and DCM ratio.
- Verification: `lesson28_buck.py` (26 checks; CCM/DCM switching simulations, motor-current ripple simulation, KERS energy accounting, scooter start dynamics with an illustrative inertia).

Lesson 29 — The step-up (boost) converter, regenerative braking and the two-quadrant chopper

Chapter V: Power Electronics (Güç Elektroniği) · Week 14 · Lesson 29 of 31

Assistant-written teaching note. ***Provenance:*** *no teacher deck for Chapter V is supplied. Built from Sen §10.4.2 (step-up chopper, printed pp. 517–519, viewer 541–543), §10.4.3 (buck–boost, printed pp. 519–521, extension), §10.4.4 and Example 10.5(b) (two-quadrant chopper, printed pp. 521–523, viewer 545–547) and §10.9 (boost power-factor correction, printed pp. 547–549). The output ripple, CCM boundary and non-ideal gain (§3.4–4) are derived here from first principles (standard results beyond Sen's text). Exam: Jan 2022 Final Q4(e) (e-scooter boost, with key). Numbers and switching simulations:* `lesson29_boost.py`.

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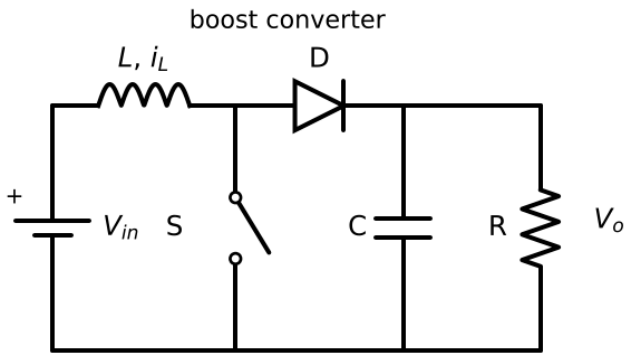
1. Place and goals

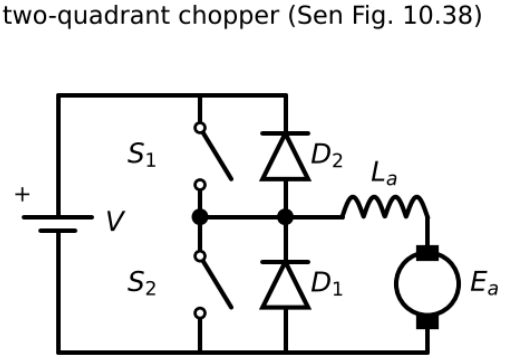
The buck converter can only lower a voltage. Many applications need the opposite: a 12 V battery feeding a 48 V bus, a PV string raised to a DC link, the scooter of Lesson 28 driven above its battery-limited speed, or a slowing motor (low back EMF) pushing energy back into a higher-voltage supply. The **boost converter** rearranges the same three components — switch, diode, inductor — to *raise* the voltage.

After this lesson you should be able to:

1. explain how storing energy in an inductor and releasing it in series with the source raises the voltage; sketch the waveforms;
2. derive $V_o = V_{in}/(1-D)$ and the current relations; compute ripple and choose L and C;
3. find the CCM boundary and explain why a boost converter must never run unloaded;
4. explain why the practical gain is limited and compute it with inductor resistance;
5. use the boost connection and the two-quadrant chopper for regenerative braking of a DC motor.

2. Circuit and operation (Sen Fig. 10.36)





The inductor is now on the **input** side; the switch S shorts the inductor's output end to ground; the diode D connects it to the output capacitor and load.

- **S on,** $0 < t < DT$**:** the inductor is connected across the source, $v_L = V_{in}$, and i_L rises linearly (Sen Eq. 10.20): energy is stored in L. The diode is reverse biased (its cathode sits at V_o, anode at 0); the capacitor alone supplies the load.
- **S off,** $DT < t < T$**:** the inductor current continues through the diode into the output. Now $v_L = V_{in} - V_o$, which must be negative for the current to fall (Sen Eq. 10.20b): the source *and* the inductor's stored energy together feed the load, at a voltage above V_{in}.

3. Steady-state relations in CCM

3.1 Voltage ratio

Volt-second balance: $V_{in}DT + (V_{in} - V_o)(1-D)T = 0$ (Sen Eqs. 10.20a,c), so

$$\boxed{V_o = \frac{V_{in}}{1-D}} \qquad \text{(Sen Eq. 10.21).} \tag{29.1}$$

$D = 0$ gives $V_o = V_{in}$ (diode always conducting); $D \to 1$ would give $V_o \to \infty$ in the ideal model — §4 shows why it does not in reality.

3.2 Currents

Power balance: $V_{in}I_{in} = V_oI_o \Rightarrow I_{in} = I_o/(1-D)$. The inductor carries the input current: $I_L = I_{in}$. The diode passes i_L only during $(1-D)T$, and its average must equal the load current: $I_{D,avg} = (1-D)I_L = I_o$ ✓ (charge balance on C). The switch carries i_L during DT: $I_{S,avg} = DI_L$. Both devices block V_o.

3.3 Inductor ripple

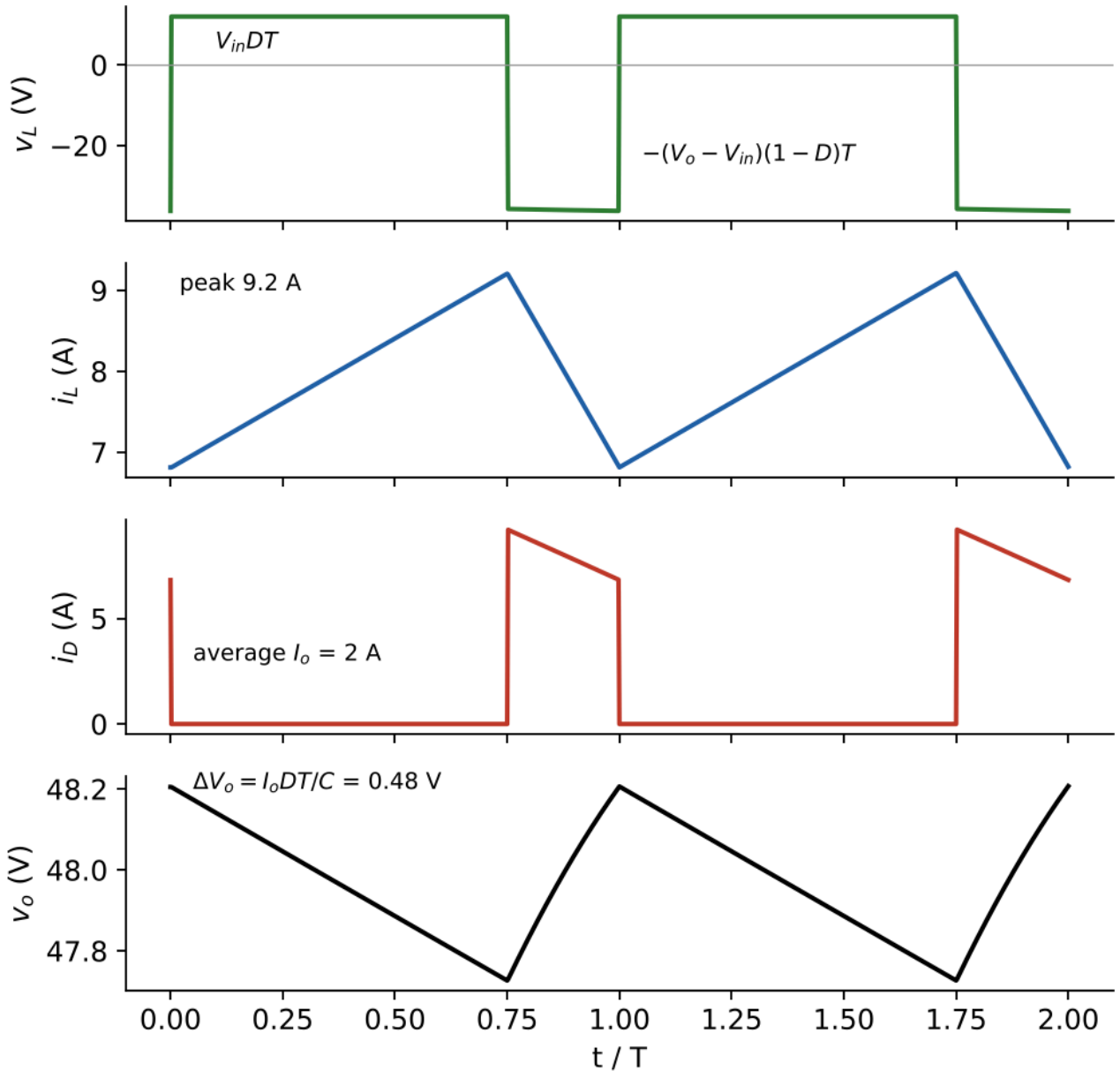
$$\Delta I_L = \frac{V_{in}DT}{L} \qquad \text{(Sen Eq. 10.20a)}, \qquad I_{L,max/min} = \frac{I_o}{1-D} \pm \frac{\Delta I_L}{2}. \tag{29.2}$$

3.4 Output ripple

During DT the diode is off and the capacitor alone supplies the (nearly constant) load current I_o:

$$\boxed{\Delta V_o = \frac{I_oDT}{C}}. \tag{29.3}$$

Unlike the buck, the boost output capacitor receives a *pulsed* current (the diode current jumps between 0 and $\approx I_L$), so its ripple-current rating is larger.



3.5 CCM boundary and the unloaded boost

The valley of i_L reaches zero when $I_L = \Delta I_L/2$, i.e. $I_o/(1-D) = V_{in}DT/(2L)$:

$$I_{o,B} = \frac{V_o T D(1-D)^2}{2L}, \qquad L_{crit} = \frac{D(1-D)^2 RT}{2}. \tag{29.4}$$

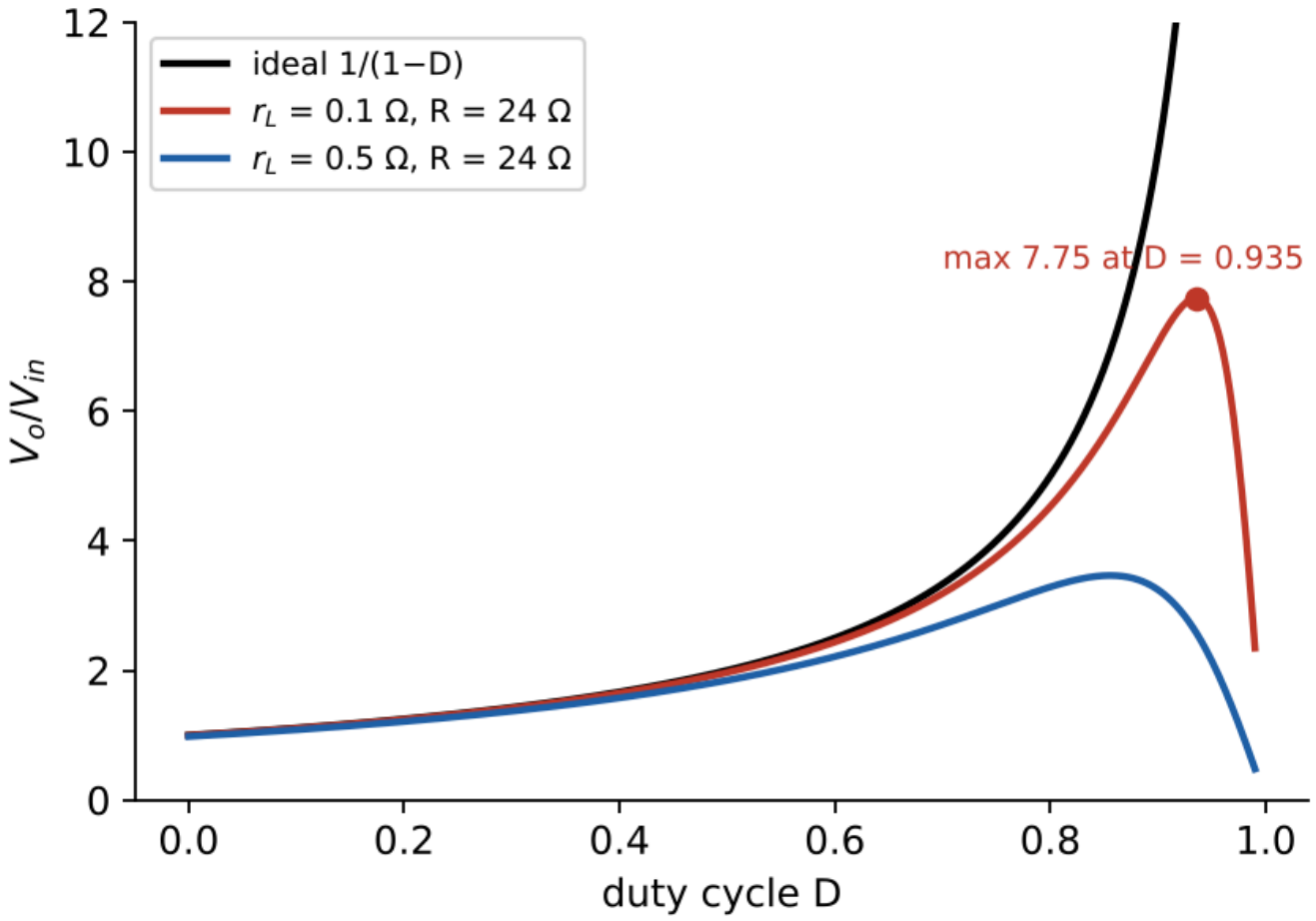
Below this load the converter enters DCM and the output voltage rises above $V_{in}/(1-D)$: by the same steps as Eq. 28.5, $V_o/V_{in} = \frac{1}{2}\left(1+\sqrt{1+4D^2/K}\right)$ with $K = 2L/(RT)$. (Example 29.1's converter at R = 400 Ω gives K = 0.019 and V_o = 72 V instead of 48 V; simulation 71.98 V.) With **no load at all**, every switching cycle pumps $\frac{1}{2}LI_{peak}^2$ into the capacitor with nowhere to go: the output voltage keeps rising until something breaks. A boost converter needs a minimum load or closed-loop control that stops switching.

4. Non-ideal gain: why $1/(1-D)$ is not reachable

Add the inductor (and switch) resistance r_L. Volt-second balance with $I_L = V_o/[R(1-D)]$:

$$V_{in} - I_L r_L - (1-D)V_o = 0 \;\Rightarrow\; \boxed{\frac{V_o}{V_{in}} = \frac{1}{1-D}\cdot\frac{1}{1+\dfrac{r_L}{R(1-D)^2}}}, \qquad \eta = \frac{1}{1+\dfrac{r_L}{R(1-D)^2}}. \tag{29.5}$$

As $D \to 1$ the input current $V_o/[R(1-D)]$ grows without limit and the $I_L^2 r_L$ loss swamps the output. The gain has a maximum at $(1-D)^2 = r_L/R$: $M_{max} = \tfrac{1}{2}\sqrt{R/r_L}$, at an efficiency of only 50 %. Practical boost converters stay below $D \approx 0.8$ (gain ≈ 4–5); larger ratios use transformers or cascades. (Derivation ours; checked by simulation.)



5. Regenerative braking and the two-quadrant chopper

5.1 The motor as the low-voltage source (Sen §10.4.2, last paragraph)

Turn Figure 29.1 (left) around: let the *DC machine* be the source V_{in} (its back EMF minus the armature drop) and the *supply* be the output V_o. With the switch across the armature and the diode toward the supply, the circuit is a boost converter that lifts the falling motor voltage up to the fixed supply voltage. Average relation (motor current I_a flowing *out* of the machine):

$$E_a - I_a R_a = (1-D)V_{supply}. \tag{29.6}$$

By adjusting D the braking current (torque) can be held constant while the speed — and E_a — fall.

5.2 Two-quadrant chopper (Sen §10.4.4, Fig. 10.38)

Combining the two connections gives a half bridge: S1 with D1 (the freewheeling diode) form a **buck** for motoring — quadrant I, $V_0 > 0$, $I_0 > 0$; S2 with D2 form a **boost** for regenerative braking — quadrant IV in Sen's (V_0, I_0) plane, $V_0 > 0$, $I_0 < 0$. The motor voltage polarity is the same in both; only the current reverses. This is the converter behind "bidirectional DC–DC converter" in the eDumper (Final 2018 Q3), traction (Final 2016 Q4) and KERS problems. A four-quadrant (reversing) drive needs a full bridge.

5.3 Application: power-factor correction (Sen §10.9)

In a computer power supply a diode bridge (Lesson 30) is followed by a boost converter whose inductor current is controlled to follow the rectified sine wave. The line current then becomes sinusoidal and in phase with the voltage (power factor ≈ 1), while the boost output is held at ≈ 400 V (Sen Figs. 10.57–10.59). Without it, a capacitor-filtered bridge draws narrow current pulses (Lesson 30 Example 30.6, PF ≈ 0.45).

6. The buck–boost converter (Sen §10.4.3 — extension)

With the inductor connected to ground and the switch and diode on either side, the inductor sees $+V_{in}$ during DT and $-V_o$ during $(1-D)T$ (output polarity inverted):

$$\frac{|V_o|}{V_{in}} = \frac{D}{1-D} \qquad \text{(Sen Eq. 10.23)},$$

below the input for $D < 0.5$ and above it for $D > 0.5$. The syllabus lists only buck and boost; this is for recognition.

7. Methods and common mistakes

1. **The inductor current is the input current** for the boost (not the load current as in the buck): $I_L = I_o/(1-D)$.
2. **Output ripple uses** DT (diode off), not $(1-D)T$: $\Delta V_o = I_o DT/C$.
3. **Switch and diode voltage stress is** V_o, not V_{in}.
4. **"Can it regenerate?"** — check the device directions. A boost from battery to motor (scooter, Jan 2022 Q4(e)) has its diode pointing toward the motor: current cannot return to the battery.
5. **Gain limits:** quoting $V_o = 12/(1-0.95) = 240$ V is unrealistic; use (29.5) when r_L is known.

8. Worked examples

Example 29.1 — Design of a 12 V → 48 V boost converter (original)

Given. V_{in} = 12 V, V_o = 48 V, P_o = 96 W (R = 24 Ω), f_s = 100 kHz; inductor ripple 30 % of I_L; output ripple 1 %. **Find** D, currents, L, C, boundary, device stresses. **Solution.** $D = 1 - 12/48 =$ **0.75**. I_o = 2 A, $I_L = I_{in} = 2/0.25 =$ **8 A**. ΔI_L = 2.4 A ⇒ $L = V_{in}DT/\Delta I_L = 12(0.75)(10^{-5})/2.4 =$ **37.5 µH**; peak current 9.2 A. ΔV_o = 0.48 V ⇒ $C = I_o DT/\Delta V_o = 2(0.75)(10^{-5})/0.48 =$ **31.25 µF**. $L_{crit} = D(1-D)^2RT/2$ = 5.6 µH; boundary load current $I_{o,B}$ = 0.3 A (R = 160 Ω). Switch: average DI_L = 6 A, peak 9.2 A, blocks 48 V; diode: average 2 A, peak 9.2 A, blocks 48 V. **Verification.** Simulation (Figure 29.2): V_o = 47.97 V, ΔI_L = 2.40 A, ΔV_o = 0.48 V, I_L = 8.01 A.

Example 29.2 — Regenerative braking with the two-quadrant chopper (Sen Example 10.5(b))

Given. As Example 28.3 (120 V, T = 4 ms, $K\Phi$ = 0.1 V/rpm, R_a = 0.2 Ω), now S2 and D2 operate at n = 350 rpm with I_0 = −100 A (ripple-free). **Find** t_{on} of S2 and the powers. **Solution.** Motor-side average voltage $V_0 = E_a + I_0 R_a = 35 - 20$ = 15 V. When S2 is on, v_0 = 0; when it is off, D2 conducts and v_0 = 120 V: $V_0 = (1 - t_{on}/T)120 \Rightarrow t_{on}/T$ = 7/8 ⇒ t_{on} = **3.5 ms** (boost relation (29.6) with V_{supply} = 120 V). Machine power $E_a I_0 = 35(-100)$ = **−3500 W** (generating); R_a loss **2000 W**; source receives $120 \times 100 \times \frac{1}{8}$ = **1500 W**. Check: 3500 = 2000 + 1500 ✓. At this low speed most of the recovered power is lost in R_a.

Example 29.3 — Doubling the scooter speed with a boost converter (Jan 2022 Final Q4(e))

Source. `exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf` p. 14. **Key confirmed.** **Given.** Motor of Example 28.6 ($K_a\Phi$ = 0.086 V·s/rad, R_a = 0.05 Ω), 48 V battery. Run at double rated speed (9550 rpm, 1000 rad/s) with half rated current (50 A). **Solution.** $E_a = 0.086 \times 1000$ = 86 V; $V_t = E_a + I_a R_a = 86 + 2.5$ = **88.5 V** > 48 V ⇒ boost. $D = 1 - 48/88.5$ = **0.458**. Battery current (ideal) $50 \times 88.5/48$ = 92 A. *Regenerative braking?* Reducing D lowers the voltage the converter imposes, but the boost diode points from the battery side to the motor: motor current cannot flow back into the battery. At most the motor briefly charges the output capacitor, raising its voltage and stopping the current — **no regeneration** with this unidirectional boost. A bidirectional (half-bridge) converter — the two-quadrant chopper of §5.2 — is needed (key and our analysis agree).

Example 29.4 — How far can a real boost converter go? (original)

Given. Example 29.1 converter with inductor-plus-switch resistance r_L = 0.1 Ω, R = 24 Ω. **Solution.** At D = 0.75: $V_o/V_{in} = 4/(1 + 0.1/(24 \times 0.0625)) = 4/1.0667$ = 3.75 ⇒ V_o = **45 V** (not 48 V), efficiency **93.8 %**; restoring 48 V needs a slightly larger D. Maximum gain at $1 - D = \sqrt{0.1/24}$ = 0.0645 (D = 0.935): $M_{max} = \frac{1}{2}\sqrt{240}$ = **7.75** (93 V from 12 V) at 50 % efficiency. Beyond that, increasing D *lowers* the output (Figure 29.3). Simulation at D = 0.75 confirms 45.0 V.

Example 29.5 — Buck–boost (extension, Sen §10.4.3)

24 V input, D = 0.6: $|V_o| = 24(0.6/0.4)$ = **36 V**, inverted polarity (simulation: 36.0 V). At D = 0.4 the same circuit gives 16 V.

9. Optional practice

P29.1. Design a boost converter from 24 V to 60 V delivering 300 W at 50 kHz with 2.5 A inductor ripple and 0.6 V output ripple.

▼ **Solution**

$D = 1 - 24/60$ = 0.6; I_o = 5 A (R = 12 Ω); I_L = 12.5 A. $L = 24(0.6)(20 \times 10^{-6})/2.5$ = **115 µH**; $C = 5(0.6)(20 \times 10^{-6})/0.6$ = **100 µF**.

P29.2 (regenerative braking). A DC motor with R_a = 0.5 Ω runs at a speed where E_a = 100 V. It brakes into a 300 V DC link through a boost connection with a constant 20 A. Find the duty cycle of the switch and the power flows.

▼ **Solution**

$E_a - I_a R_a = 90\ \text{V} = (1-D)300 \Rightarrow D = \mathbf{0.7}$. Mechanical power converted $E_a I_a$ = 2000 W; R_a loss 200 W; returned to the link 90×20 = **1800 W**. As the speed falls, E_a falls and D must increase to keep 20 A.

P29.3 (concept). Why is a boost converter dangerous to operate without load, while a buck converter is not?

▼ **Solution**

In a buck converter, removing the load just lets the output rise toward V_{in} (DCM), never above it. In a boost converter each cycle transfers $\frac{1}{2}LI^2$ from the inductor to the output capacitor regardless of load; with no load the capacitor voltage keeps rising each cycle until the capacitor or switch fails (§3.5).

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10. Summary

Quantity (CCM, ideal)	Boost
Voltage ratio	$V_o = V_{in}/(1-D)$
Currents	$I_L = I_{in} = I_o/(1-D)$; diode avg I_o; switch avg DI_L
Inductor ripple	$\Delta I_L = V_{in}DT/L$
Output ripple	$\Delta V_o = I_o DT/C$
CCM boundary	$L_{crit} = D(1-D)^2 RT/2$
Non-ideal gain	$M = \frac{1}{1-D}\frac{1}{1+r_L/[R(1-D)^2]}$, $M_{max} = \frac{1}{2}\sqrt{R/r_L}$
Regenerative braking	$E_a - I_a R_a = (1-D)V_{supply}$ (two-quadrant chopper)
Buck–boost (extension)	$\lvert V_o\rvert = DV_{in}/(1-D)$, inverted

11. Vocabulary (EN–TR)

English	Türkçe
step-up (boost) converter	yükseltici (boost) dönüştürücü
buck–boost converter	düşürücü–yükseltici dönüştürücü
two-quadrant chopper	iki bölgeli kıyıcı
regenerative braking	rejeneratif (geri kazanımlı) frenleme
power-factor correction (PFC)	güç faktörü düzeltme
DC link	DA bara (DA ara devre)
voltage gain	gerilim kazancı
voltage stress	gerilim zorlanması
bidirectional converter	çift yönlü dönüştürücü
half bridge / full bridge	yarım köprü / tam köprü

12. Sources

- Sen 3rd ed.: §10.4.2, Eqs. 10.20–10.21, Fig. 10.36 (printed pp. 517–519, viewer 541–543); §10.4.3, Eqs. 10.22–10.23, Fig. 10.37 (printed pp. 519–521, viewer 543–545); §10.4.4, Fig. 10.38, Example 10.5(b) (printed pp. 521–523, viewer 545–547); §10.9, Figs. 10.56–10.59 (printed pp. 547–549, viewer 571–573).

- Jan 2022 Final Q4(e): `exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf` p. 14 (key).
- Derived here (standard, beyond Sen's text): Eq. 29.3 output ripple, Eq. 29.4 boundary, Eq. 29.5 non-ideal gain.
- Verification: `lesson29_boost.py` (12 checks; boost, non-ideal boost and buck–boost switching simulations, Sen Example 10.5(b) energy balance).

Lesson 30 — Single-phase diode rectifiers

Chapter V: Power Electronics (Güç Elektroniği) · Week 14 · Lesson 30 of 31

*Assistant-written teaching note. **Provenance:** no teacher deck for Chapter V is supplied. Sen treats diode rectifiers inside its controlled-rectifier section: §10.2.1 single-phase circuits, Eq. 10.1 and Fig. 10.17 (half-wave diode, R load), the full converter Eq. 10.3 and Example 10.1 (a diode bridge behaves as the full converter at α = 0), Problem 10.1 (ripple factor), Problem 10.5 (battery charging) and §10.9, Fig. 10.58a (capacitor-filtered bridge) — printed pp. 492–502 and 547–549, viewer 516–526 and 571–573. The RL-load extinction-angle analysis, the capacitor-filter ripple estimates and the input-current harmonic analysis are derived here from first principles (standard results). Numbers and simulations:* `lesson30_single_phase_rectifiers.py`.

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1. Place and goals

A rectifier (doğrultucu) converts AC to DC. Nearly every converter connected to the grid starts with one: a battery charger, a laptop adapter, the DC link of a motor drive, the front end of a computer supply (Sen §10.9). Diode rectifiers are *uncontrolled*: the output depends only on the supply and the load. They are the simplest power converters and the reference against which controlled rectifiers (thyristors, Sen §10.2) are described.

After this lesson you should be able to:

1. explain which diodes conduct and when, for half-wave and full-wave (centre-tap and bridge) rectifiers;
2. compute the average and rms output voltages, ripple factor, form factor, diode currents and peak inverse voltage for resistive loads;
3. analyse inductive loads — the half-wave RL extinction angle, the freewheeling diode and the “constant current” (highly inductive) bridge with its square-wave line current, harmonics and power factor;
4. estimate the ripple, conduction angle and peak current of a capacitor-filtered bridge and explain its poor input power factor;
5. connect the rectifier to the machines of Chapter IV: DC motor fed from a bridge, battery charging.

2. Ideal-diode rules

An ideal diode conducts (zero voltage) when its anode would otherwise be more positive than its cathode, and blocks (zero current) otherwise. In rectifiers:

- with a **resistive** load, a diode conducts exactly while the supply drives it forward;
- with an **inductive** load, the inductor keeps current flowing after the supply reverses, so a diode can conduct with the supply *negative* (until the current reaches zero, or until another diode takes over);
- with a **capacitive** load (or a battery), diodes conduct only while the supply exceeds the capacitor (battery) voltage — short pulses near the peaks.

Notation: supply $v_s = \sqrt{2}\,V \sin\omega t = V_m \sin\omega t$ (V = rms, Sen writes V_p). Peak inverse voltage (PIV) = the largest reverse voltage a diode must block.

3. Half-wave rectifier

3.1 Resistive load (Sen §10.2.1, Fig. 10.17)

The diode conducts for $0 < \omega t < \pi$: $v_o = v_s$; for $\pi < \omega t < 2\pi$ it blocks: $v_o = 0$.

$$V_0 = \frac{1}{2\pi}\int_0^{\pi} V_m \sin\theta \, d\theta = \frac{V_m}{\pi} = \frac{\sqrt{2}V}{\pi} = 0.450\,V \quad \text{(Sen Eq. 10.1)}, \qquad V_{rms} = \frac{V_m}{2}. \tag{30.1}$$

Figures of merit (Sen Problem 10.1 defines the ripple factor):

$$\text{form factor } FF = \frac{V_{rms}}{V_0} = \frac{\pi}{2} = 1.571, \qquad \text{ripple factor } RF = \frac{\sqrt{V_{rms}^2 - V_0^2}}{V_0} = \sqrt{FF^2 - 1} = 1.211 \tag{30.2}$$

RF > 1: the AC content of the output exceeds its DC value — a poor DC source. Ripple frequency = supply frequency. PIV = V_m. The supply current has a DC component, which saturates supply transformers (Lesson 15: DC magnetizes the core) — one more reason half-wave rectifiers are used only at very low power.

3.2 RL load: the extinction angle

With an inductor in the load the current lags. While the diode conducts, $L\,di/dt + Ri = V_m \sin\omega t$ with $i(0) = 0$; the solution is the steady sinusoid plus a decaying transient:

$$i(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \varphi) + \sin\varphi \; e^{-\omega t/\tan\varphi}\right], \qquad Z = \sqrt{R^2 + (\omega L)^2},\ \tan\varphi = \frac{\omega L}{R}. \tag{30.3}$$

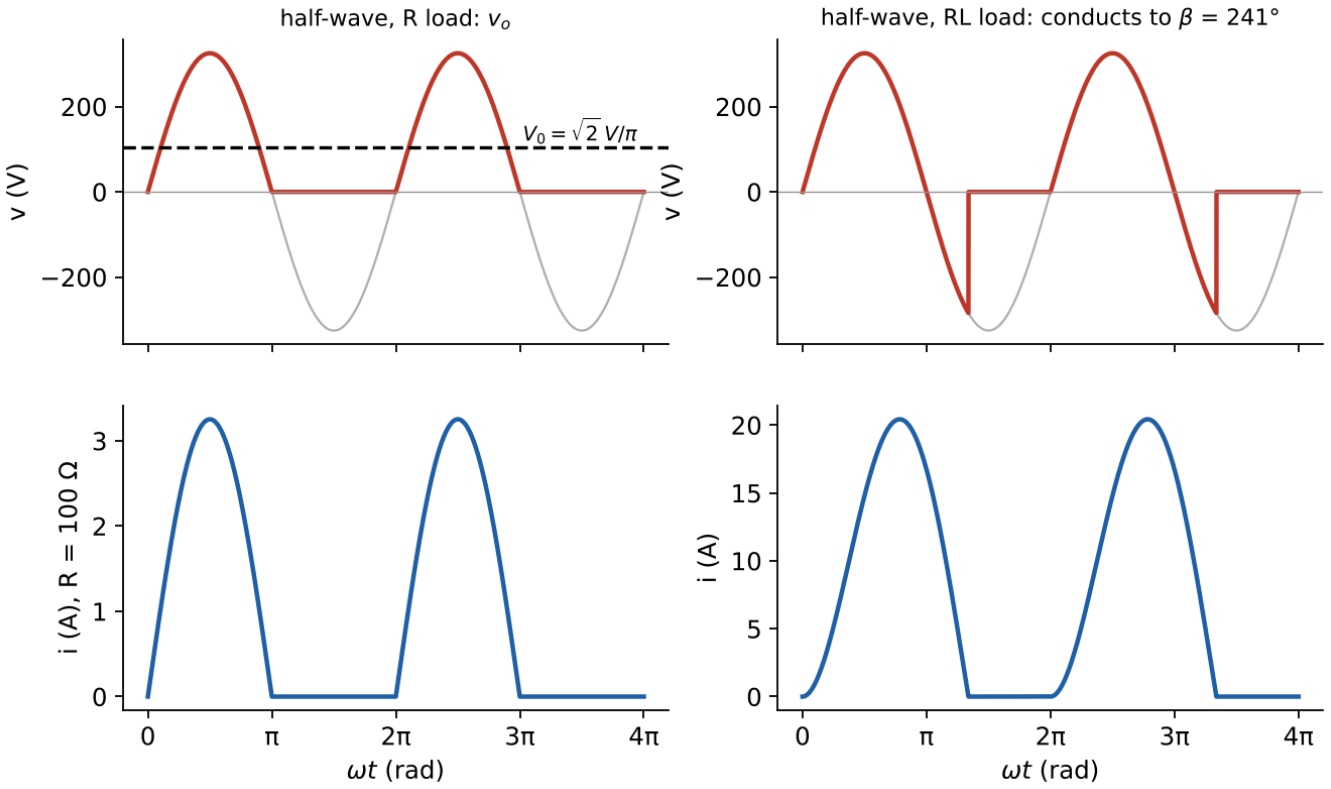
The current reaches zero — and the diode turns off — at the **extinction angle** β > π, found from $i(\beta) = 0$ numerically. Between π and β the output voltage follows the *negative* supply (Sen Fig. 10.19 shows the same effect for a thyristor), so

$$V_0 = \frac{V_m}{2\pi}(1 - \cos\beta) < \frac{V_m}{\pi}, \qquad I_0 = \frac{V_0}{R} \tag{30.4}$$

(the inductor has no average voltage). The more inductance, the larger β and the lower V_0.

3.3 Freewheeling diode

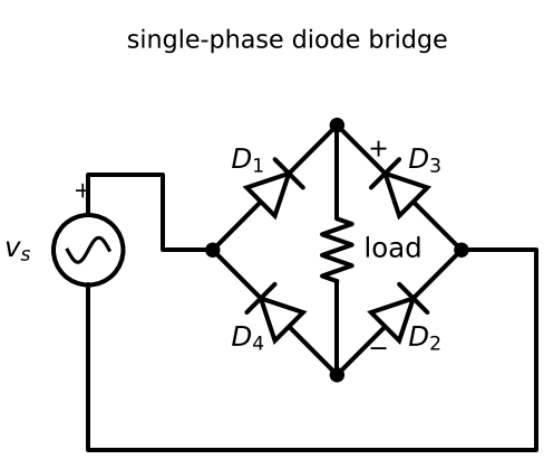
A second diode D_{FW} across the load (anode at the negative terminal) changes the picture: as soon as v_s goes negative, D_{FW} conducts, clamps v_o to zero and carries the inductor current; the main diode turns off. Now v_o is the same as for a resistive load, $V_0 = V_m/\pi$ regardless of L, and with large L the load current is nearly constant, $I_0 = V_0/R$, carried by the main diode in positive half-cycles and by D_{FW} in negative ones (each ≈ $I_0/2$ average). The same freewheeling action appears in the semiconverter (Sen Fig. 10.24) and in the buck converter (Lesson 28).

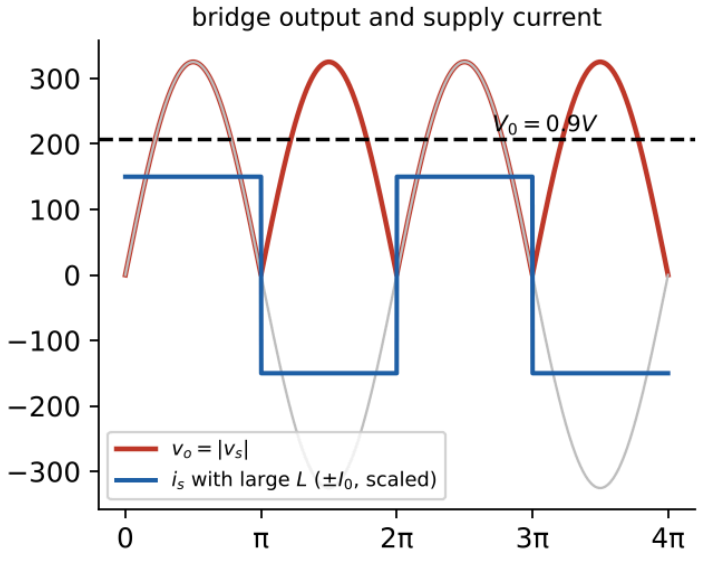


4. Full-wave rectifiers

4.1 Centre-tapped transformer and bridge

- **Centre-tap:** a transformer with a centre-tapped secondary feeds two diodes; each half-winding conducts in its half-cycle. Two diodes, but each must block $2V_m$ and the transformer is poorly utilized.
- **Bridge (Figure 30.2):** four diodes, no centre tap. In the positive half-cycle D_1 and D_2 conduct (the supply's upper terminal → D_1 → load + … load − → D_2 → lower terminal); in the negative half-cycle D_3 and D_4. Two diodes always conduct in series; each blocks V_m. A diode bridge is the thyristor full converter of Sen Fig. 10.20 with every device fired at α = 0.





4.2 Resistive load

$v_o = |v_s|$:

$$V_0 = \frac{2V_m}{\pi} = \frac{2\sqrt{2}}{\pi}V = 0.900\,V, \qquad V_{rms} = V, \qquad RF = \sqrt{\frac{\pi^2}{8} - 1} = 0.483, \tag{30.5}$$

ripple at $2f$, PIV V_m (bridge) or $2V_m$ (centre-tap). Equation (30.5) is Sen Eq. 10.3 with α = 0.

4.3 Highly inductive load (constant current)

With a large series inductance (e.g. a DC motor armature, Sen Fig. 10.21) the load current is essentially constant, $i_o = I_0$. The output voltage is still $|v_s|$ (two diodes always conduct), so $V_0 = 0.9V$ and $V_0 = E_a + I_0 R_a$ for a motor (Sen Eq. 10.4). The **supply current** is a square wave of amplitude I_0, in phase with v_s:

$$I_{s,rms} = I_0, \qquad I_{s1,rms} = \frac{2\sqrt{2}}{\pi}I_0 = 0.900\,I_0, \qquad \text{THD} = \sqrt{\frac{I_s^2 - I_{s1}^2}{I_{s1}^2}} = 48.3\,\%.$$

The fundamental is in phase with the voltage (displacement factor 1), but the harmonics carry no average power, so

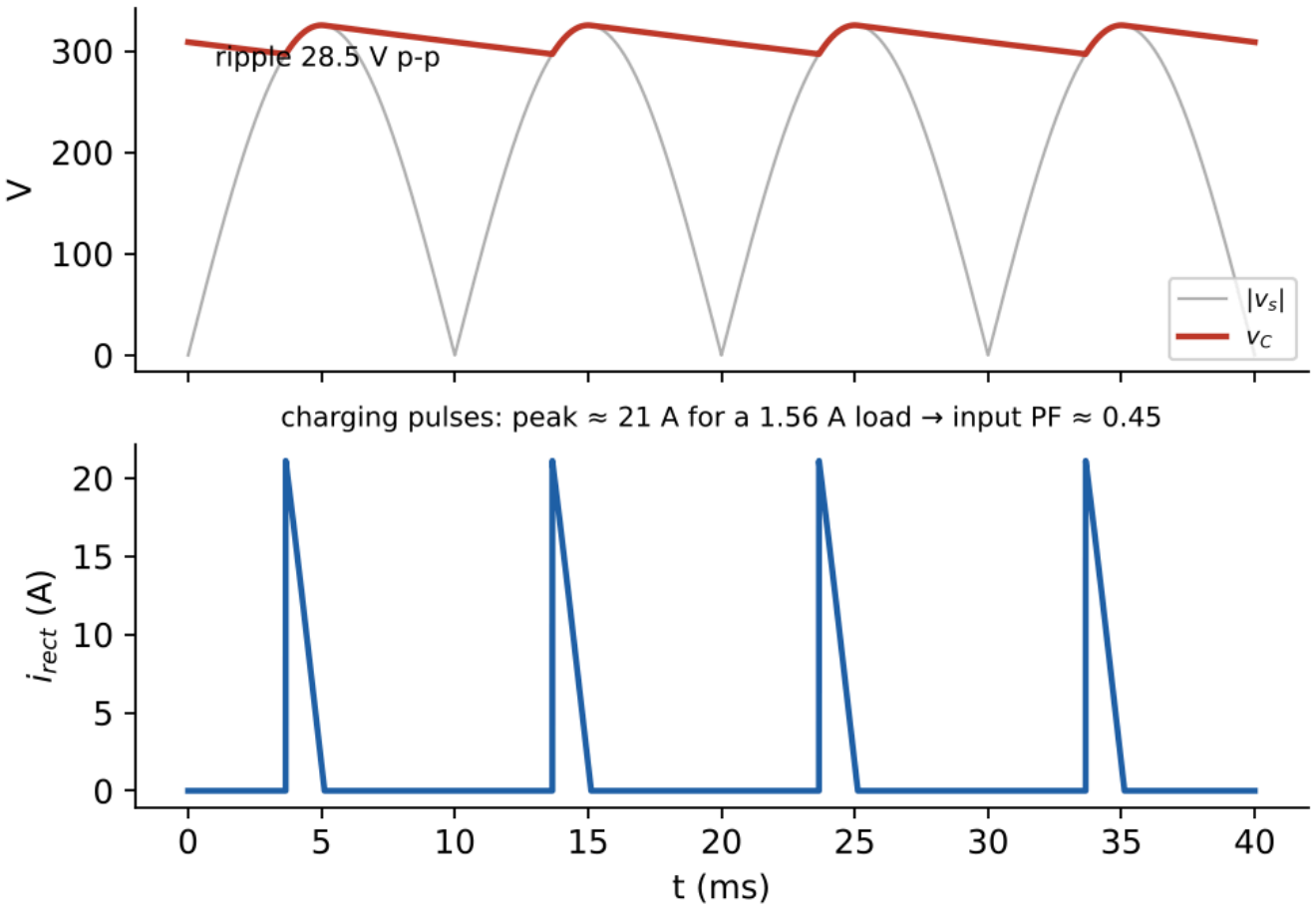
$$PF = \frac{P}{S} = \frac{V I_{s1}\cos 0}{V I_s} = \frac{2\sqrt{2}}{\pi} = 0.900. \tag{30.6}$$

Sen Example 10.1(c) computes the same way ($PF = P_s/S$ with an rms square-wave current) for a firing angle of 19.2°.

5. Capacitor-filtered bridge (Sen §10.9, Fig. 10.58a)

Electronic loads need a smooth voltage, so a large capacitor is placed across the bridge output. The capacitor charges to nearly V_m at each peak, then supplies the load alone while the diodes are reverse biased, discharging roughly exponentially ($v_C = V_m e^{-t/RC}$), until $|v_s|$ rises above v_C again.

- **Ripple, simple estimate:** if the ripple is small, the load current $I_0 \approx V_m/R$ is supplied for almost a half period: $\Delta V \approx \dfrac{I_0}{2fC} = \dfrac{V_m}{2fRC}$. This *overestimates* the ripple because the actual discharge interval is shorter (half period minus the conduction time); a better estimate is $\Delta V = V_m(1 - e^{-t_{off}/RC})$ with $t_{off} = T/2 - t_{cond}$.
- **Conduction:** the diodes conduct only for a short interval near each peak; the whole charge drawn by the load in a half cycle must be delivered in that interval, so the **peak current is many times the average**.
- **Input power factor:** the line current is a train of narrow pulses: rich in harmonics, PF typically 0.4–0.7 — the motivation for power-factor-correction boost stages (Sen §10.9, Lesson 29 §5.3).



6. Methods and common mistakes

1. **Average vs rms.** DC meters, motor speed and battery charge depend on the *average*; heating of resistors depends on the *rms*. A half-wave rectifier delivers V_{rms}^2/R to a resistor, not V_0^2/R.
2. **Inductive load without freewheeling diode:** the output goes negative after π; do not use V_m/π.
3. **PF of a rectifier is not cos φ.** With non-sinusoidal current, PF = (displacement factor) × (distortion factor I_1/I). A diode bridge with constant current has displacement factor 1 but PF 0.9.
4. **PIV:** bridge diodes block V_m; centre-tap diodes block $2V_m$.
5. **Capacitor filter:** $\Delta V = I/(2fC)$ is a quick, conservative estimate; check that the ripple is small before trusting it.

7. Worked examples

Example 30.1 — Half-wave rectifier with a resistor (Sen Eq. 10.1 and Problem 10.1(a))

Given. 230 V, 50 Hz; R = 100 Ω. **Find** V_0, I_0, V_{rms}, load power, RF, PIV, diode currents. **Solution.** V_m = 325.3 V. $V_0 = V_m/\pi$ = **103.5 V**, I_0 = 1.035 A. $V_{rms} = V_m/2$ = **162.6 V** ⇒ $P = V_{rms}^2/R$ = **264.5 W** (exactly half of the 529 W a direct AC connection would give). RF = **1.21**, FF = 1.571. PIV = **325 V**. Diode: average 1.035 A, rms 1.63 A.

Example 30.2 — Half-wave rectifier with an RL load (original)

Given. 230 V, 50 Hz; R = 10 Ω, L = 50 mH. **Find** the extinction angle, V_0 and I_0. **Solution.** $X = \omega L$ = 15.71 Ω, Z = 18.62 Ω, φ = 57.5°. Solving $\sin(\beta-\varphi)+\sin\varphi\, e^{-\beta/\tan\varphi}=0$: β = **240.8°**. $V_0 = \frac{325.3}{2\pi}(1-\cos 240.8^\circ)$ = **77.0 V** (vs 103.5 V for the R load), I_0 = **7.70 A**. **Verification.** Direct simulation of the diode–RL circuit: average current 7.70 A, current zero at 240.8°.

Example 30.3 — Adding a freewheeling diode (original)

Given. Same supply; R = 10 Ω, L = 1 H (highly inductive), with a freewheeling diode. **Solution.** $V_0 = V_m/\pi$ = **103.5 V** ⇒ I_0 = **10.35 A**, nearly constant (simulated ripple 1.1 A p-p at L = 1 H). Main diode and D_{FW} each carry ≈ 5.2 A average. Compared with Example 30.2 the freewheeling diode raises the average output by removing the negative part of v_o.

Example 30.4 — Bridge with a resistive load (original)

Given. 230 V, 50 Hz, bridge, resistive load. **Find** V_0, V_{rms}, RF, PIV for bridge and centre-tap versions. **Solution.** $V_0 = 0.9\times 230$ = **207.1 V**; V_{rms} = 230 V; RF = **0.483** (vs 1.21 half-wave); ripple at 100 Hz. PIV: bridge **325 V**; centre-tap (each half-winding 230 V) **651 V**.

Example 30.5 — DC motor on a diode bridge (Sen Example 10.1 machine, α = 0)

Given. Sen's motor: $K\Phi$ = 0.09 V/rpm, R_a = 0.4 Ω, large L_a (ripple-free current); single-phase 120 V, 60 Hz supply through a **diode** bridge; load current 30 A. **Find** speed, supply apparent power, power factor, fundamental current, THD. **Solution.** $V_0 = 0.9\times 120$ = **108.0 V**; $E_a = 108 - 30(0.4)$ = 96.0 V ⇒ N = 96/0.09 = **1067 rpm**. Supply current: ±30 A square wave ⇒ I_s = 30 A, S = 3600 VA; $P = V_0 I_0$ = 3241 W; **PF = 0.900**; I_{s1} = 27.0 A (in phase), THD = **48.3 %**. **Comparison with Sen.** Sen's thyristor converter must deliver 102 V for 1000 rpm, which needs α = 19.2° ($108\cos 19.2^\circ$ = 102 V ✓). A diode bridge cannot adjust the speed — that is why thyristors (or a chopper after the bridge, Lesson 28) are used.

Example 30.6 — Capacitor-filtered bridge (original; Sen Fig. 10.58a)

Given. 230 V, 50 Hz, bridge, C = 470 µF, load 200 Ω (≈ 500 W). **Find** DC voltage, ripple, conduction interval, peak diode current, input power factor; the C needed for 5 % ripple. **Solution (ideal source and diodes, exact simulation).** Average V_{dc} = **311.6 V**; ripple **28.5 V p-p** (8.7 %); the simple estimate $V_m/(2fRC)$ = 34.6 V is conservative (the refined estimate with $t_{off} = T/2 - t_{cond}$ agrees within 3 %). The diodes conduct only ≈ 26° per half-cycle (52° per cycle); peak charging current ≈ **21 A** for an average load current of 1.56 A. Line current: rms 4.71 A, fundamental 2.11 A ⇒ **PF ≈ 0.45**. For 5 % ripple the simple formula gives $C = V_m/(2fR\cdot 0.05V_m)$ = **1 mF** (on the safe side). **Interpretation.** Smooth DC, but a very peaky input current — the price of the simple capacitor filter.

Example 30.7 — Charging a battery through one diode (original variant of Sen Problem 10.5)

Given. 120 V, 60 Hz; battery 10 × 7.2 V = 72 V with total resistance 0.6 Ω; a single diode (no thyristor). **Find** the conduction interval, average charging current, charging power and resistor loss. **Solution.** The diode conducts while $V_m\sin\theta > E$: from $\theta_1 = \sin^{-1}(72/169.7)$ = **25.1°** to θ_2 = 154.9°. Average current

$$I_0 = \frac{1}{2\pi R}\left[2V_m\cos\theta_1 - E(\theta_2-\theta_1)\right] = \mathbf{38.3\ A},$$

peak $(169.7 - 72)/0.6$ = 163 A. Charging power EI_0 = **2.75 kW**; resistor loss $I_{rms}^2 R$ = $70.3^2 \times 0.6$ = **2.96 kW** — more than the useful power. A series inductor or current-controlled charger is needed in practice; Sen's version uses a thyristor fired at 45° to limit the current.

8. Optional practice

P30.1. A half-wave rectifier on 120 V, 60 Hz feeds 20 Ω. Find V_0, I_0, load power and PIV.

▼ **Solution**
$V_0 = 169.7/\pi$ = 54.0 V; I_0 = 2.70 A; $P = (V_m/2)^2/R = 84.85^2/20$ = 360 W; PIV 169.7 V.

P30.2. A diode bridge on 230 V supplies a highly inductive load drawing 10 A. Find V_0, the load resistance, supply apparent power, real power and power factor.

▼ **Solution**
V_0 = 207.1 V; $R = V_0/I_0$ = 20.7 Ω; $S = 230 \times 10$ = 2300 VA (square-wave current, rms 10 A); $P = 207.1 \times 10$ = 2071 W; PF = 0.900.

P30.3. A 9 V rms transformer and bridge must supply 1 A DC with at most 5 % ripple (50 Hz). Estimate C.

▼ **Solution**
V_m = 12.7 V (ignoring diode drops); ΔV = 0.64 V; $C \approx I/(2f\Delta V) = 1/(100 \times 0.64)$ ≈ **15.7 mF**. (Real designs subtract two diode drops, ≈ 1.6 V, from V_m.)

P30.4 (concept). Why is the supply current of a half-wave rectifier undesirable for a supply transformer, while a bridge is fine?

▼ **Solution**
The half-wave current has a DC component (I_0 flows in one direction only). In the transformer it adds a DC MMF that biases the core toward saturation (Lessons 15–16), increasing magnetizing current and losses. The bridge draws a symmetric (zero-average) current.

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9. Summary

Circuit / load	V_0	Other results
Half-wave, R	$V_m/\pi = 0.45V$	$V_{rms} = V_m/2$, RF 1.21, PIV V_m, ripple at f
Half-wave, RL	$\frac{V_m}{2\pi}(1 - \cos\beta)$	β from $i(\beta) = 0$ (Eq. 30.3)
Half-wave + freewheeling diode	V_m/π	current continuous with large L
Bridge (or centre-tap), R	$2V_m/\pi = 0.9V$	$V_{rms} = V$, RF 0.483, ripple at 2f, PIV V_m ($2V_m$ c.t.)
Bridge, constant current	$0.9V$	$I_s = I_0$, $I_{s1} = 0.9I_0$, THD 48.3 %, PF 0.9
Bridge + capacitor	$\approx V_m$	$\Delta V \lesssim V_m/(2fRC)$, narrow current pulses, low PF

10. Vocabulary (EN–TR)

English	Türkçe
rectifier / half-wave / full-wave	doğrultucu / yarım dalga / tam dalga
bridge rectifier	köprü doğrultucu
centre-tapped transformer	orta uçlu transformatör
peak inverse voltage (PIV)	tepe ters gerilim
ripple factor / form factor	dalgalanma faktörü / biçim faktörü
extinction angle	sönme açısı
freewheeling diode	serbest geçiş diyodu
smoothing (filter) capacitor	süzgeç (filtre) kondansatörü
conduction angle	iletim açısı
total harmonic distortion (THD)	toplam harmonik bozulma
displacement / distortion factor	kayma / bozulma faktörü
battery charging	akü şarjı

11. Sources

- Sen 3rd ed.: §10.2.1 single-phase circuits, Eq. 10.1, Figs. 10.17–10.21, Eq. 10.3–10.4, semiconverter and freewheeling diode (Fig. 10.24), Example 10.1 (printed pp. 492–502, viewer 516–526); Problems 10.1 and 10.5 (printed pp. 551–552, viewer 575–576); §10.9, Fig. 10.58a (printed pp. 547–549, viewer 571–573).
- Derived here (standard): Eq. 30.3 RL current and extinction angle, Eq. 30.6 power factor with harmonics, capacitor-filter estimates.
- Verification: `lesson30_single_phase_rectifiers.py` (16 checks: numerical averages/rms, extinction angle by root-finding and simulation, freewheeling simulation, Fourier analysis of the square-wave current, exact ideal capacitor-filter simulation with power balance).

Lesson 31 — Three-phase diode rectifiers

Chapter V: Power Electronics (Güç Elektroniği) · Week 14 · Lesson 31 of 31

Assistant-written teaching note. ***Provenance:*** *no teacher deck for Chapter V is supplied. Built from Sen §10.2.2 (three-phase circuits, printed pp. 502–511, viewer 526–535): half-wave diode rectifier (Eq. 10.9, Fig. 10.25), full converter (Eq. 10.10, Fig. 10.27) at firing angle α = 0, Examples 10.2 and 10.3 (data reused at α = 0). The ripple, line-current harmonic and power-factor results are derived here (standard Fourier analysis). Numbers:* `lesson31_three_phase_rectifiers.py`.

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1. Place and goals

Above a few kilowatts rectifiers are fed from the three-phase grid (Sen §10.2.2): industrial DC drives, electrolysis, EV fast chargers, the DC link of every large inverter drive. Three phases give a far smoother DC output and a better input current than one phase, with the same diode rules as Lesson 30.

After this lesson you should be able to:

1. identify which diode conducts at each instant in the three-phase half-wave (3-pulse) and bridge (6-pulse) rectifiers;
2. derive $V_0 = \frac{3\sqrt{6}}{2\pi}V_p$ (half-wave) and $V_0 = \frac{3\sqrt{2}}{\pi}V_{LL} = 1.35V_{LL}$ (bridge), the ripple range and frequency, and the PIV;
3. draw the line current for a ripple-free DC current and compute its rms value, fundamental, harmonics ($n = 6k \pm 1$), THD and power factor $3/\pi$;
4. apply the bridge to DC-motor and charger problems and compare single- and three-phase rectifiers.

Notation: phase (line-to-neutral) rms voltage V_p (Sen's notation), line-to-line $V_{LL} = \sqrt{3}V_p$, $v_{AN} = \sqrt{2}V_p \sin\omega t$, v_{BN} lagging by 120°, v_{CN} leading by 120° (Sen Eqs. 10.6–10.8; Chapter I three-phase conventions).

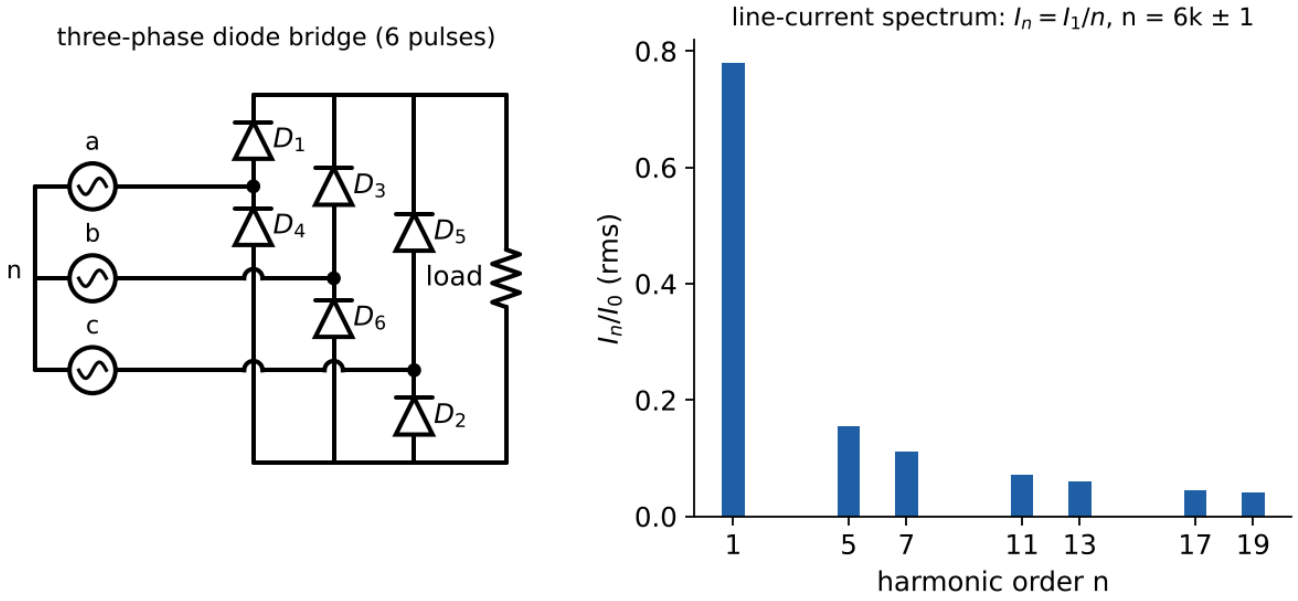
2. Three-phase half-wave (three-pulse) rectifier (Sen Fig. 10.25)

Three diodes with anodes on phases a, b, c and cathodes joined at the positive output; the load returns to the neutral. **The diode whose anode is at the highest instantaneous voltage conducts** (Sen p. 502); the other two are reverse biased by line-to-line voltages. So v_o is the **upper envelope** of the phase voltages. D_1 conducts for 30° < ωt < 150°, then D_2, then D_3 — 120° each.

$$V_0 = \frac{1}{2\pi/3}\int_{30^\circ}^{150^\circ} \sqrt{2}V_p \sin\theta\, d\theta = \frac{3\sqrt{6}}{2\pi}V_p = 1.170\,V_p \qquad \text{(Sen Eq. 10.9)}. \tag{31.1}$$

Ripple: three pulses per cycle (180 Hz on 60 Hz), v_o between $V_m/2$ (at the crossover points, $V_m \sin 30^\circ$) and V_m; RF = 0.183. Each diode must block the line-to-line peak $\sqrt{3}V_m = \sqrt{2}V_{LL}$. The supply currents contain DC components (each phase conducts in one direction only), so this circuit is mainly a stepping stone to the bridge.

3. Three-phase bridge (six-pulse) rectifier



Six diodes: an upper group D_1, D_3, D_5 (cathodes joined, positive rail) and a lower group D_4, D_6, D_2 (anodes joined, negative rail), numbered in conduction order as in Sen Fig. 10.27. At every instant **one upper diode (the most positive phase) and one lower diode (the most negative phase) conduct**:

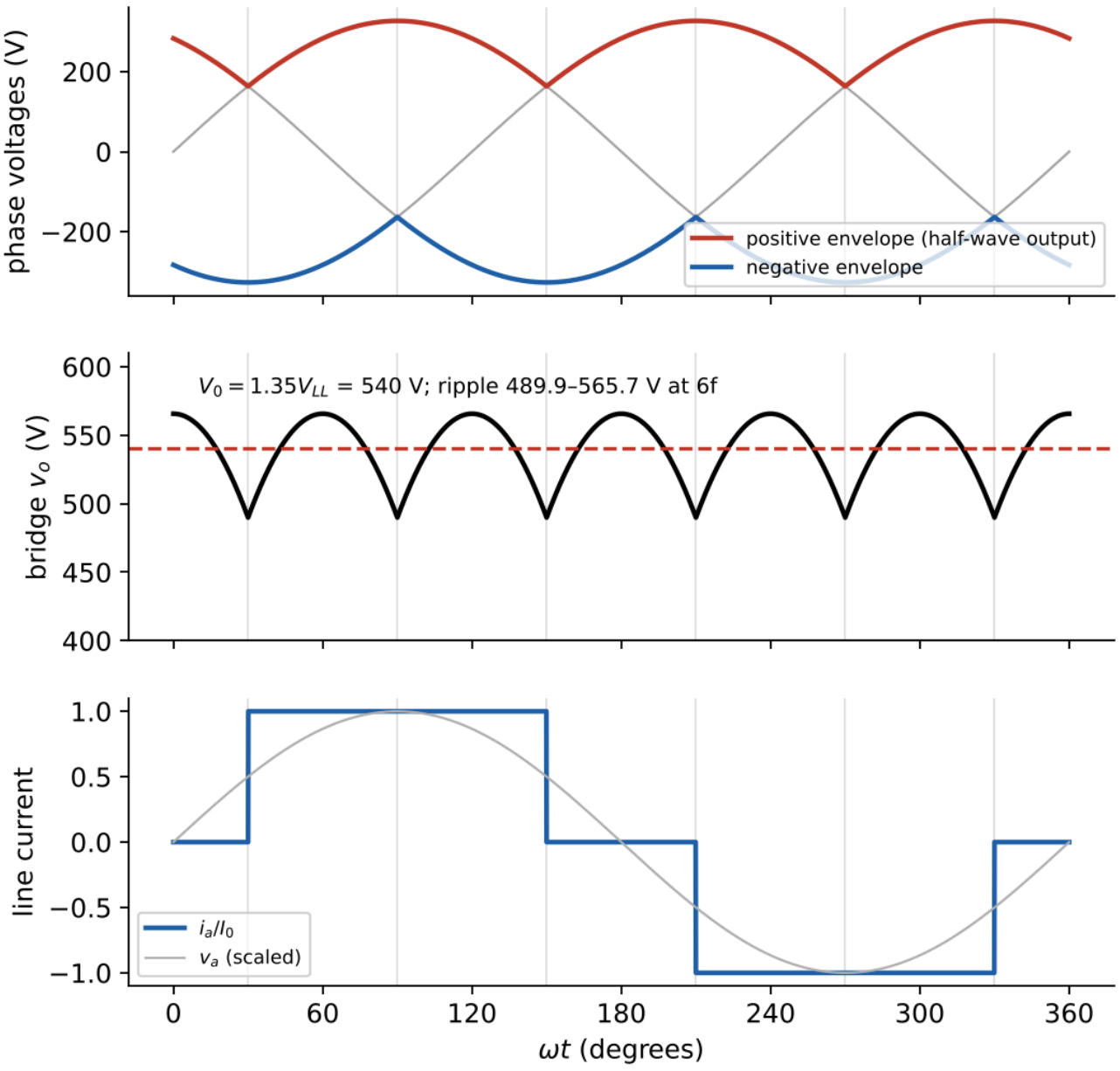
$$v_o = \max(v_a, v_b, v_c) - \min(v_a, v_b, v_c),$$

the distance between the upper and lower envelopes — always one of the six line-to-line voltages near its peak. The conducting pair changes every 60°: (1,6) v_{ab} → (1,2) v_{ac} → (3,2) v_{bc} → (3,4) v_{ba} → (5,4) v_{ca} → (5,6) v_{cb} → … Each diode conducts for 120°.

Over one 60° interval $v_o = v_{ab} = \sqrt{2}V_{LL}\sin(\theta + 30^\circ)$ for 30° < θ < 90°:

$$\boxed{V_0 = \frac{1}{\pi/3}\int_{60^\circ}^{120^\circ} \sqrt{2}V_{LL}\sin\phi\, d\phi = \frac{3\sqrt{2}}{\pi}V_{LL} = \frac{3\sqrt{6}}{\pi}V_p = 1.350\,V_{LL}} \qquad \text{(Sen Eq. 10.10 at } \alpha = 0\text{)}. \quad (31.2)$$

Ripple: six pulses per cycle (300 Hz on 50 Hz), v_o between $\sqrt{2}V_{LL}\cos 30^\circ = 1.225V_{LL}$ and $\sqrt{2}V_{LL} = 1.414V_{LL}$; RF = 0.042 — already a good DC voltage without any filter. PIV = $\sqrt{2}V_{LL}$.



4. Line current with a ripple-free DC current

With a large DC-side inductance (motor armature, DC-link choke) the output current is a constant I_0 (Sen's assumption in Fig. 10.27). Phase a carries $+I_0$ while D_1 conducts (120°, centred on the positive peak of v_a), $-I_0$ while D_4 conducts (120°, centred on the negative peak), and zero for two 60° gaps: a **quasi-square wave** (Sen p. 507).

- rms value: $I_s = I_0\sqrt{\frac{240^\circ}{360^\circ}} = \sqrt{\frac{2}{3}}\,I_0 = 0.816\,I_0$ (Sen Example 10.3: 106.1 A for 130 A);
- Fourier series (the wave is centred on the voltage peaks, so the fundamental is **in phase** with v_a):

$$I_{s1} = \frac{\sqrt{6}}{\pi} I_0 = 0.780\,I_0, \qquad I_{sn} = \frac{I_{s1}}{n} \ \text{for } n = 6k \pm 1 = 5, 7, 11, 13, \ldots \tag{31.3}$$

  (no even harmonics, no triplen harmonics);
- THD = $\sqrt{(I_s/I_{s1})^2 - 1}$ = **31.1 %** (vs 48.3 % for the single-phase bridge);
- power factor

$$PF = \frac{3V_p I_{s1}\cos 0}{3V_p I_s} = \frac{3}{\pi} = 0.955; \tag{31.4}$$

- power check: $3V_pI_{s1} = \sqrt{3}V_{LL}\,\frac{\sqrt{6}}{\pi}I_0 = \frac{3\sqrt{2}}{\pi}V_{LL}I_0 = V_0I_0$ ✓ — the harmonics carry no average power.

With a controlled (thyristor) bridge the fundamental would lag by α and $PF = (3/\pi)\cos\alpha$ (Sen Example 10.3: 0.68 at α = 44.4°): a diode bridge has the best power factor of the family.

5. Single- versus three-phase

	1-φ bridge (230 V)	**3-φ bridge (400 V)**
V_0	$0.9V$ = 207 V	$1.35V_{LL}$ = 540 V
ripple frequency	2f = 100 Hz	6f = 300 Hz
ripple factor (R load)	0.483	0.042
line-current THD (constant I_0)	48.3 %	31.1 %
power factor (constant I_0)	0.900	0.955
PIV	V_m	$\sqrt{2}V_{LL}$

6. Methods and common mistakes

1. **Phase vs line voltage.** Sen's V_p is the phase rms voltage: $V_0 = 2.34V_p = 1.35V_{LL}$. Using V_{LL} in $3\sqrt{6}V_p/\pi$ gives a value √3 too large.
2. **Half-wave vs bridge:** 1.17 V_p vs 2.34 V_p — the bridge adds the negative envelope.
3. **Which diodes conduct?** Upper group: most positive phase; lower group: most negative phase. Never two diodes of the same group.
4. **Harmonics:** only $6k \pm 1$; the 5th is 20 % of the fundamental. Do not include 3rd harmonics for a three-wire bridge.
5. **PF ≠ 1 even though the fundamental is in phase**: distortion factor $I_{s1}/I_s = 3/\pi$.

7. Worked examples

Example 31.1 — Three-phase half-wave rectifier (Sen Eq. 10.9; Example 10.2 data)

Given. V_p = 120 V, 60 Hz, resistive load. **Find** V_0, ripple range and frequency, conduction interval, PIV. **Solution.** $V_0 = 1.170 \times 120$ = **140.3 V**; v_o between 84.9 V and 169.7 V at **180 Hz**; each diode conducts **120°**; RF = 0.18; PIV = $\sqrt{3} \times 169.7$ = **294 V**. (Check against Sen Example 10.2: with thyristors fired at α = 60°, $V_0 = 140.3\cos 60^\circ$ = 70.2 V — Sen's answer.)

Example 31.2 — Three-phase bridge on the 400 V grid (original)

Given. 400 V line-to-line, 50 Hz, diode bridge. **Find** V_0, ripple range, RF, PIV. **Solution.** $V_0 = 1.3505 \times 400$ = **540.2 V**; v_o between $565.7\cos 30^\circ$ = **489.9 V** and **565.7 V** (75.8 V p-p) at **300 Hz**; RF = **0.042**; PIV = **566 V** (choose ≥ 1200 V diodes in practice for transients).

Example 31.3 — Line current, harmonics and power factor (original)

Given. The bridge of Example 31.2 supplying a ripple-free I_0 = 100 A. **Find** the line current rms, fundamental, the first harmonics, THD, PF, diode currents. **Solution.** $I_s = 0.8165 \times 100$ = **81.6 A**; $I_{s1} = 0.7797 \times 100$ = **78.0 A**, in phase with v_{an}; I_5 = 15.6 A, I_7 = 11.1 A, I_{11} = 7.1 A, I_{13} = 6.0 A;

THD **31.1 %**; PF **0.955**. Power: $\sqrt{3}(400)(78.0)$ = 54.0 kW = $V_0 I_0$ = 540.2 × 100 ✓. Each diode: average $I_0/3$ = 33.3 A, rms $I_0/\sqrt{3}$ = 57.7 A. (Numerical Fourier analysis in the verifier confirms every value; Figure 31.1.)

Example 31.4 — Sen Example 10.3's motor on a diode bridge

Given. 3-φ 480 V, 60 Hz; motor R_a = 0.1 Ω, $K\Phi$ = 0.3 V/rpm, rated current 130 A (ripple-free). **Find** the speed on a diode bridge and compare with Sen's thyristor converter. **Solution.** $V_0 = 1.35 \times 480$ = **648.2 V** (Sen's "648 cos α" at α = 0). $E_a = 648.2 - 13$ = 635.2 V ⇒ **2117 rpm** — fixed by the supply. To run at Sen's 1500 rpm the converter must give 463 V: $\cos\alpha = 463/648.2$ ⇒ α = **44.4°** (Sen's answer), at the cost of PF 0.68 instead of 0.955 (supply current rms 106.1 A in both cases). Diode bridge + chopper (Lesson 28) is the modern alternative: the bridge keeps PF 0.955 and the chopper sets the motor voltage.

Example 31.5 — Front end of an EV fast charger (original)

Given. 400 V, 50 Hz grid → diode bridge (large DC choke) → buck converter → 400 V battery charged at 50 kW. Converter losses neglected. **Find** the DC-link voltage, buck duty cycle, DC-link current, battery current, line current. **Solution.** V_{dc} = **540.2 V**; buck $D = 400/540.2$ = **0.740**; battery current 125 A; DC-link current $50\,000/540.2$ = **92.6 A**; line current rms 0.8165×92.6 = **75.6 A** (fundamental 72.2 A, PF 0.955). Power check $\sqrt{3} \times 400 \times 72.2$ = 50.0 kW ✓.

8. Optional practice

P31.1. A three-phase half-wave diode rectifier is supplied from a 230/400 V system (230 V phase). Find V_0.

▼ **Solution**
$V_0 = 1.170 \times 230$ = **269 V**.

P31.2. A diode bridge is fed from a 690 V (line-to-line) wind-turbine bus. Find V_0 and the PIV.

▼ **Solution**
$V_0 = 1.35 \times 690$ = **932 V**; PIV $\sqrt{2} \times 690$ = **976 V**.

P31.3. A bridge carries a ripple-free 50 A. Find the average and rms diode currents and the line-current rms value.

▼ **Solution**
Diode: average 16.7 A, rms 28.9 A. Line current rms $\sqrt{2/3} \times 50$ = 40.8 A.

P31.4 (concept). Why does the three-phase bridge line current contain no 3rd harmonic?

▼ **Solution**
The quasi-square wave has half-wave symmetry (no even harmonics) and its 60° gaps and 120° blocks are arranged so that the 3rd-harmonic components cancel: $b_3 \propto \cos(3 \times 30^\circ) = 0$. More generally, the three line currents are identical waves shifted by 120°; triplen harmonics would be in phase in all three lines and need a neutral path, which the bridge does not have.

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9. Summary

Quantity	3-pulse (half-wave)	6-pulse (bridge)
V_0	$\frac{3\sqrt{6}}{2\pi} V_p = 1.17 V_p$	$\frac{3\sqrt{2}}{\pi} V_{LL} = 1.35 V_{LL}$

Quantity	3-pulse (half-wave)	6-pulse (bridge)
conduction per diode	120°	120° (pairs change every 60°)
ripple frequency	3f	6f
output range	$V_m/2$ … V_m	$1.225V_{LL}$ … $1.414V_{LL}$
PIV	$\sqrt{2}V_{LL}$	$\sqrt{2}V_{LL}$
line current (constant I_0)	(contains DC)	rms $0.816I_0$, $I_1 = 0.780I_0$, $I_n = I_1/n$, n = 6k±1
PF	—	$3/\pi = 0.955$

10. Vocabulary (EN–TR)

English	Türkçe
three-phase bridge rectifier	üç fazlı köprü doğrultucu
six-pulse / three-pulse	altı darbeli / üç darbeli
upper / lower diode group	üst / alt diyot grubu
envelope	zarf
line-to-line voltage	hat (fazlar arası) gerilimi
quasi-square wave	yarı kare dalga
harmonic order	harmonik derecesi
DC link / DC choke	DA ara devre / DA şok bobini
fast charger	hızlı şarj cihazı

11. Sources

- Sen 3rd ed.: §10.2.2, Eqs. 10.6–10.10, Figs. 10.25–10.28, Examples 10.2–10.3 (printed pp. 502–511, viewer 526–535).
- Derived here (standard): ripple range, Fourier series of the line current (Eq. 31.3), PF = 3/π (Eq. 31.4), comparison table.
- Verification: `lesson31_three_phase_rectifiers.py` (16 checks: numerical envelopes and averages, conduction intervals, Fourier harmonics and power balance, Sen Example 10.2–10.3 cross-checks).